Chapter 2

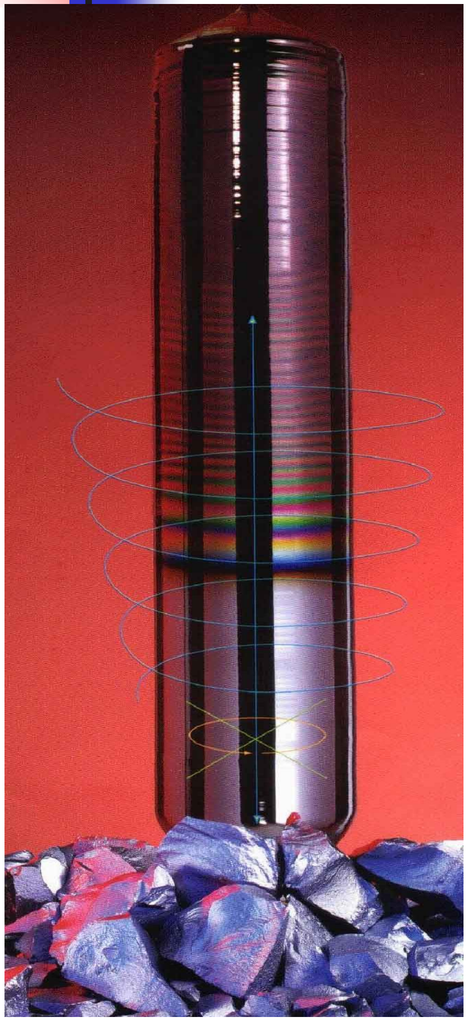
Chip Level Interconnection

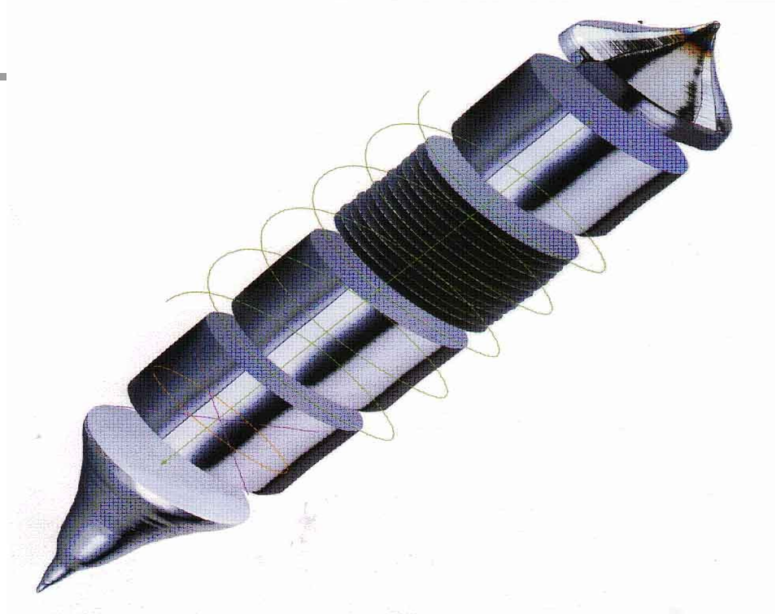
芯片互连技术

宁 宁

典型的IC封装工艺流程

1. Wafer In
2. Wafer Grinding (WG研磨)
3. Wafer Saw (WS 切割)
4. Die Attach (DA 黏晶)
5. Epoxy Curing (EC 银胶烘烤)
6. Wire Bond (WB 引线键合)
7. Die Coating (DC 晶粒封胶/涂覆)
8. Molding (MD 塑封)
9. Post Mold Cure (PMC 模塑后烘烤)
10. Dejunk/Trim (DT 去胶去纬)
11. Solder Plating (SP 锡铅电镀)
12. Top Mark (TM 正面印码)
13. Forming/Singular (FS 去框/成型)
14. Lead Scan (LS 检测)
15. Packing (PK 包装)





- 电子级硅所含的硅的纯度很高，可达 **99.9999 99999 %**
- 中德电子材料公司制作的晶棒(长度达一公尺，重量超过一百公斤)

Wafer Back Grinding

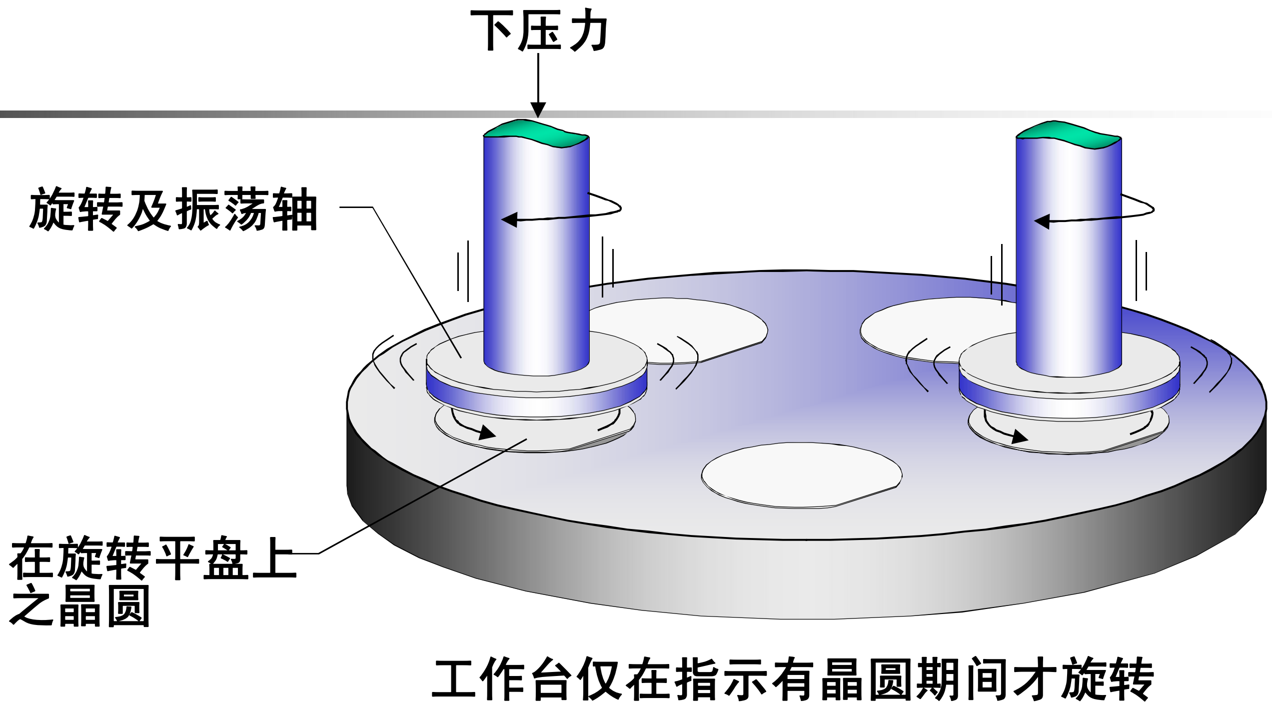
- **Purpose**

 The wafer backgrind process reduces the thickness of the wafer produced by silicon fabrication (FAB) plant. The wash station integrated into the same machine is used to wash away debris left over from the grinding process.

- **Process Methods:**

  1) Coarse grinding by mechanical.（粗磨）

  2) Fine polishing by mechanical or plasma etching. (细磨抛光)

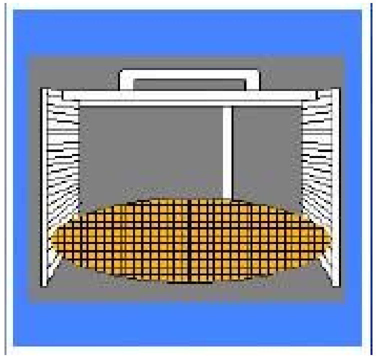


■ **Method:**

The wafer is first mounted on a **backgrind tape** and is then loaded to the backgrind machine **coarse wheel**. As the coarse grinding is completed, the wafer is transferred to a **fine wheel for polishing**.

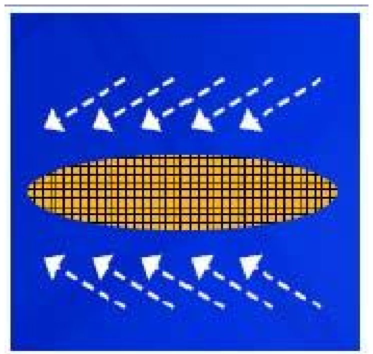
Wafer Back Grinding process

1. Load and Align



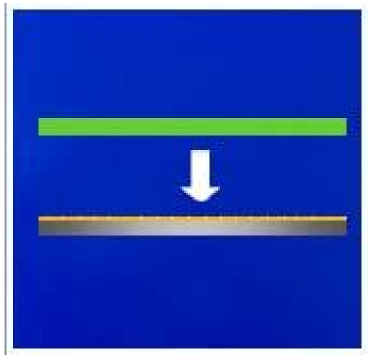
Objective:
To load and align the wafer into the wafer cleaning and tape lamination machine.

2. Wafer cleaning



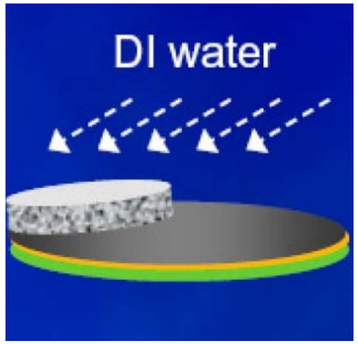
Objective:
To clean the wafer for the next lamination step.

3. Back grind Tape lamination



Objective:
To laminate a protective layer of film on the circuitry surface of the wafer .

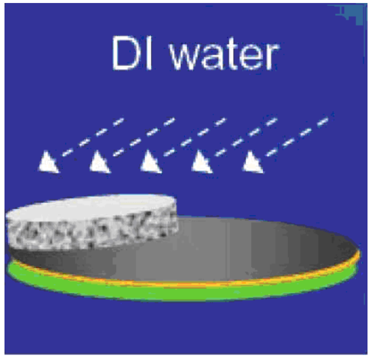
4. Coarse grinding



Objective:
To reduce the thickness
with a coarse grinding wheel.

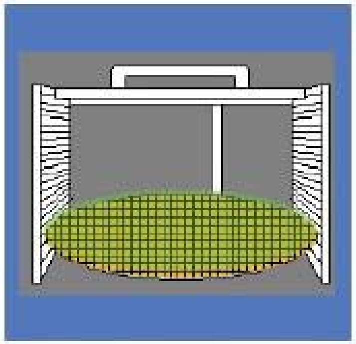
Wafer Back Grinding process (Cont.)

5. Fine polishing



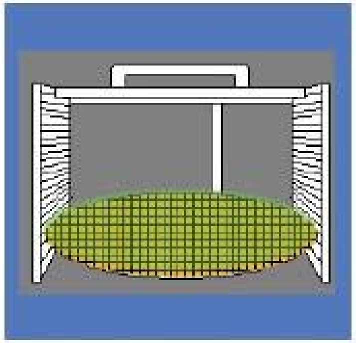
Objective:
To eliminate or reduce the stress built up during coarse grinding and polish the surface to remove micro cracks with a fine grinding wheel.

6. Unload



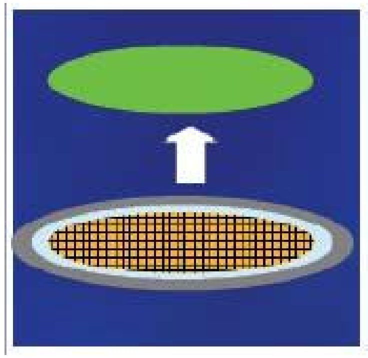
Objective:
To unload the wafer from back grinding machine.

7. Load



Objective:
To load the wafer to wafer mounter.

8. Tape removal



Objective:
To remove the back grind tape after wafer mounted on the frame.

Wafer Back Grinding Issues and Challenges

Issues

- Ease of process
  - –Thin wafer handling from one step to another
  - –Back grinding tape removal
  - –Excessive stresses removal or reduction from the wafer.（应力）
- Yield
  - –Wafer breakage due to stress built up during thinning process.
  - –Scratches.（划痕）
  - –Die metallization smearing.（污点，模糊）
- Equipment stability and capability

Challenges

- **Market requirements drive for very thin wafer (<3 mils)**
- **Flip chip wafer back grinding**

Wafer sawing

■ Wafer Separation Process

▸ **Purpose:**

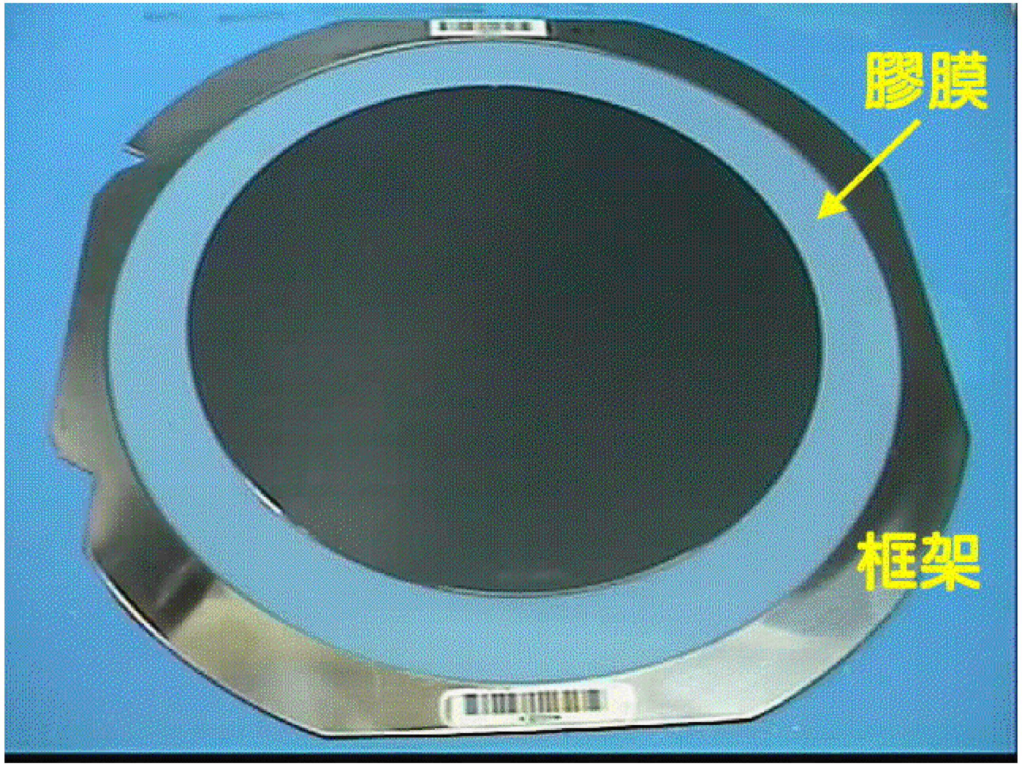
The wafer separation process is to divide the wafer into individual dice or chips.

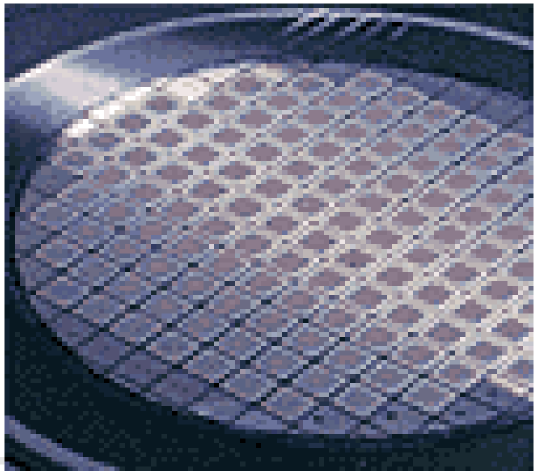
Process Methods:

1) **Sawing (with diamond-impregnated saw blade) 锯切**
   - ☐ Single or dual cut
   - ☐ Step cut or bevel cut

2) **Partial scribing** (with laser beam, diamond-tipped scribing tool, or diamond-impregnated saw blade) 局部划片器

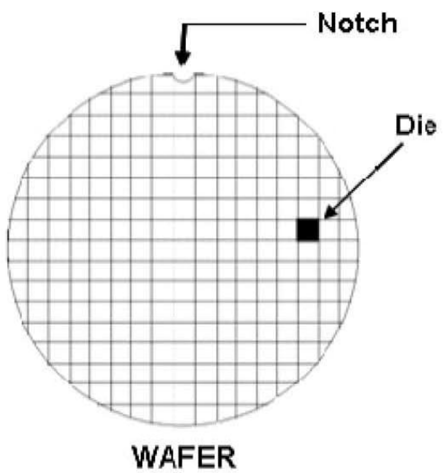
Wafer sawing



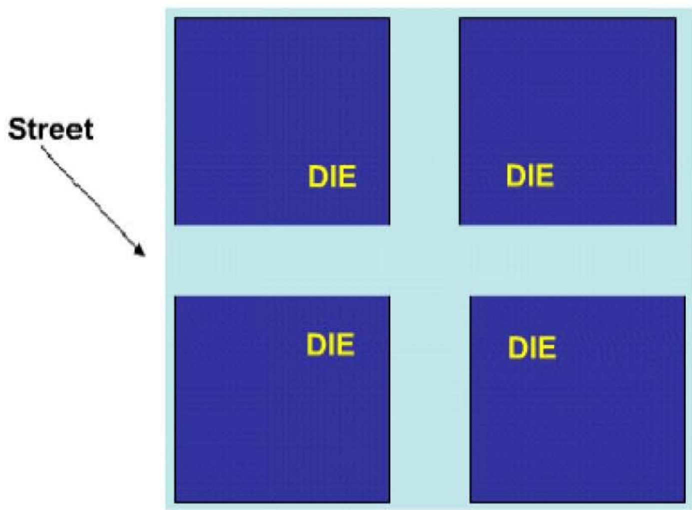


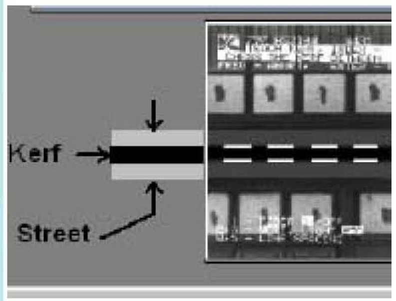
Wafer sawing

- Wafer Sawing is a Front-of-Line (FOL) operation that cuts the wafer along the streets separating the individual die. Streets, also called **scribe lines**, are lines on the wafer that separate each individual die from the surrounding dice. **Kerf width** is the saw width. After the wafer is sawn, the wash station, using a detergent, removes residual cut material from the wafer.

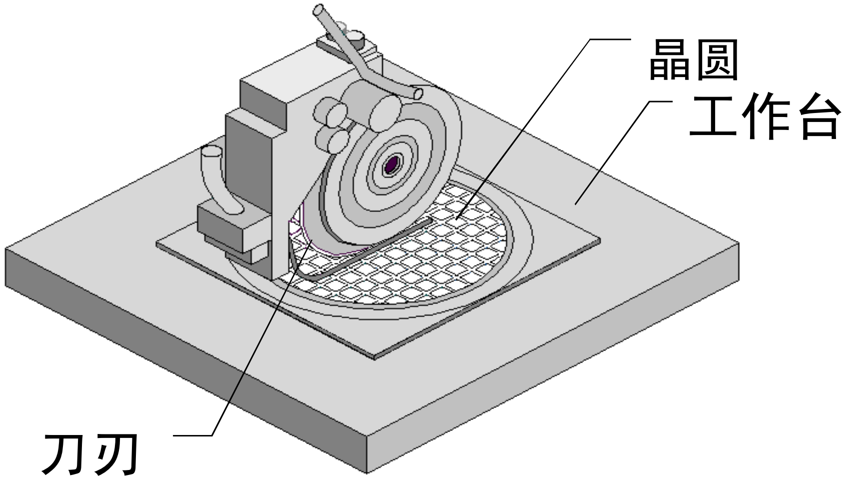


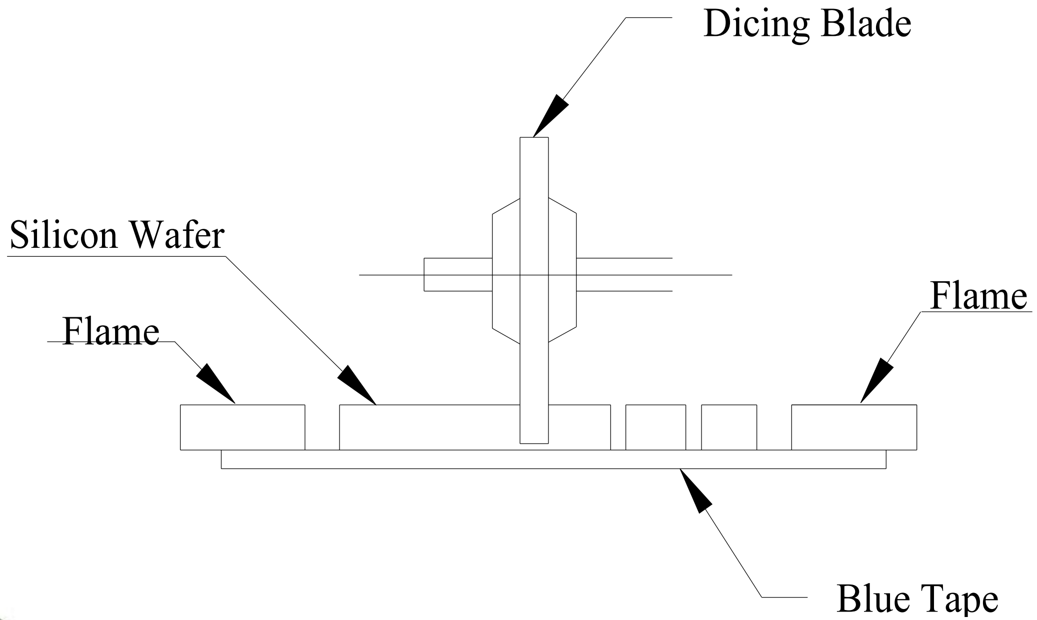
Wafer Parts

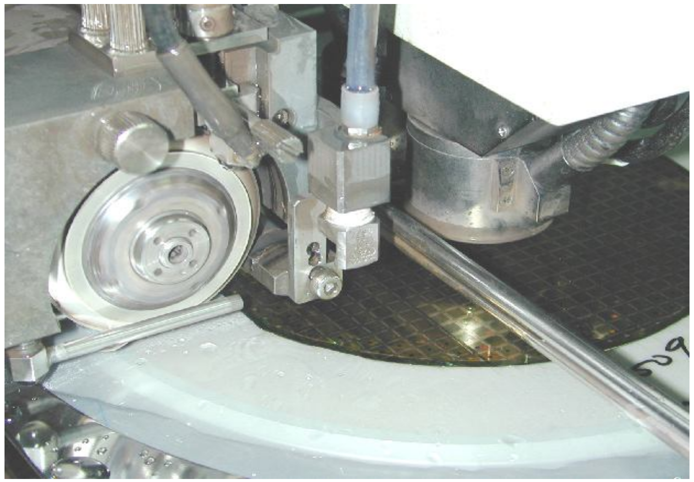




切割设备示意图





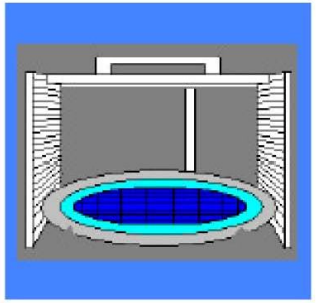


- 两次进刀切割法

Wafer sawing

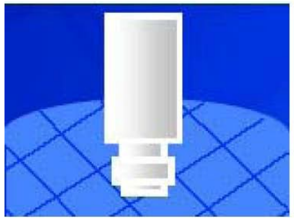
- The SAWING process is broken down into four steps:

1. Load and Align



Objective:
To load and align the mounted wafer into the cutting area in consistent placement.

2. Pattern Recognition System (PRS)

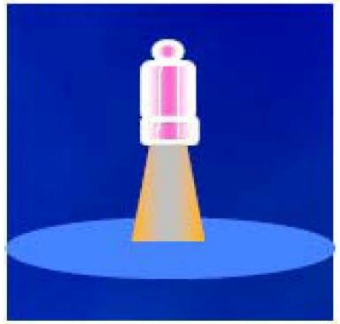


Objective:
To align the theta (rotation) position of the wafer, so that the saw blades cut parallel to the streets.

3. Cut

Objective:
To separate dice from a wafer with resin-bonded **diamond wheel**. (**First blade** is used to remove metal structures and stresses on street for **second blade.**)

4. Wash, Rinse, Dry and Unload



Objective:
To **rinse slurry (silicon dust)** before it dries with de-ionized water and CO2. Also to dry wafer by pinning and with **clean air**, and unload wafer.

Wafer Sawing Issues and Challenges

- Issues:
  - Ease of process
    - --Die **chipping** control （碎屑）
    - --Multiple die types and sizes processing
  - Yield
    - --Saw on die
    - --Scratches （划痕）
    - --Chipping
    - --Die crack
  - Equipment stability and capability

- Challenges:
  - Smaller kerf width for more die per wafer
  - Larger wafer size (300mm) with multiple die types and sizes

Wire Bonding Technology
-- Die Attach Process

□ Purpose:

The die attach process is to attach the sawed die in the right orientation accurately onto the substrate with a bonding medium in between to enable the next wire bond first level interconnection operation .

□ Process Methods

1)Semi-automated

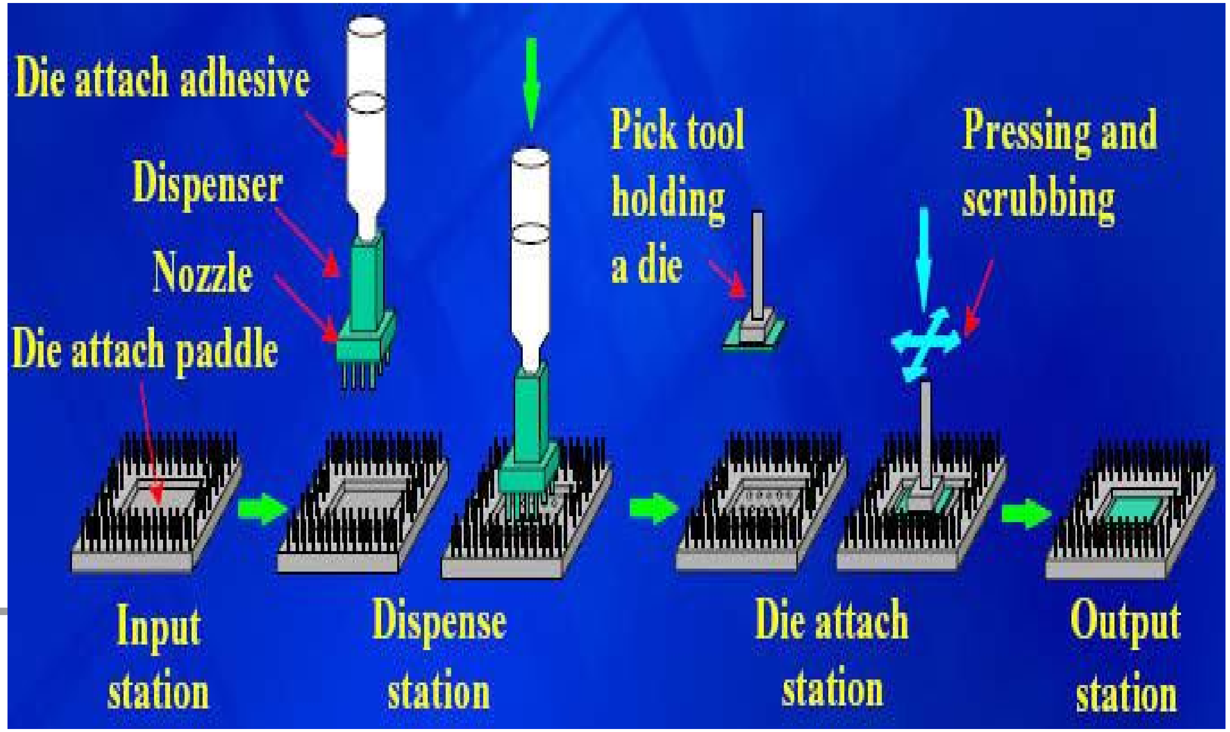
eutectic die attach.

低共熔物芯片粘接

2)Fully automated

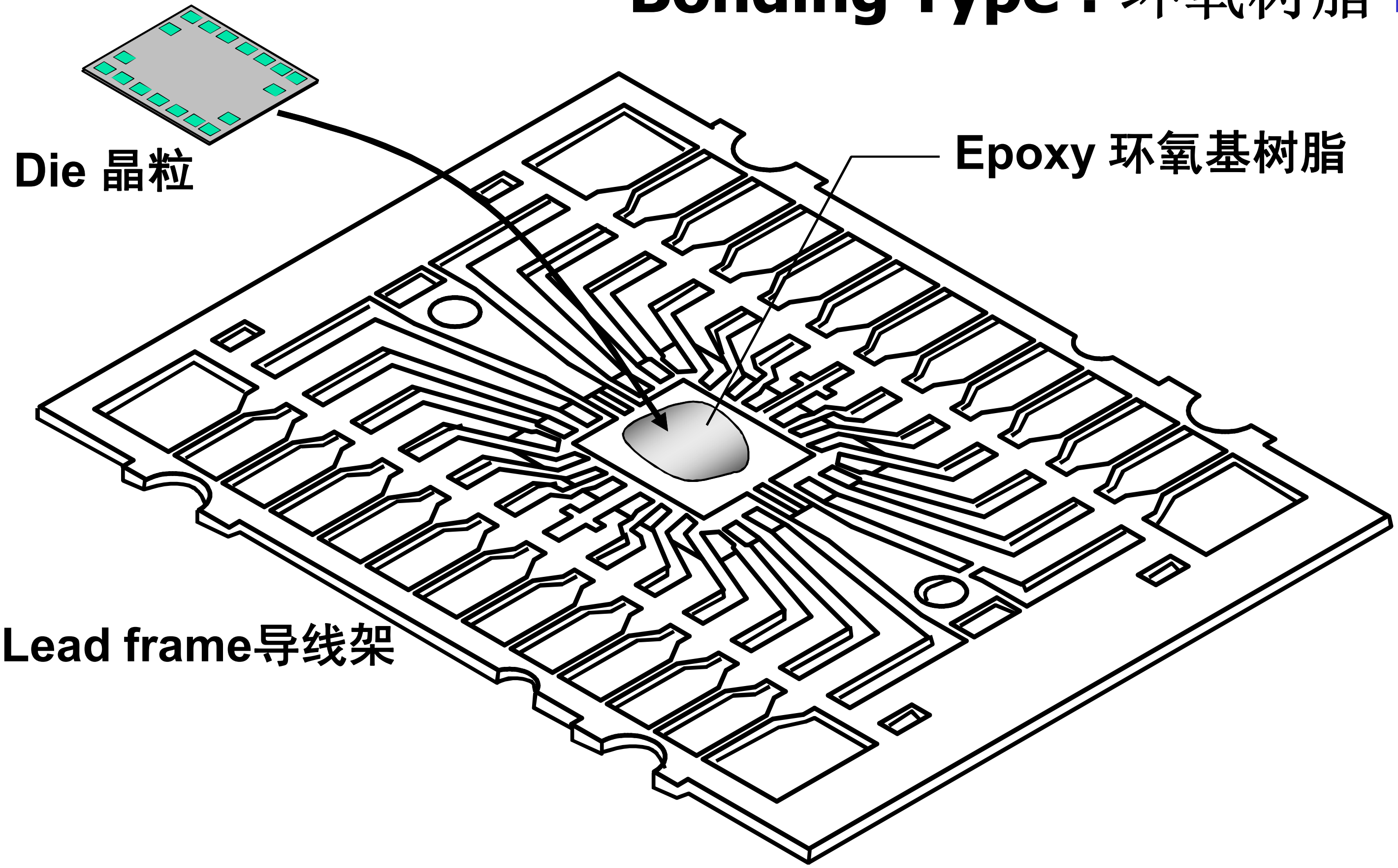
adhesive die attach.

胶粘剂粘接



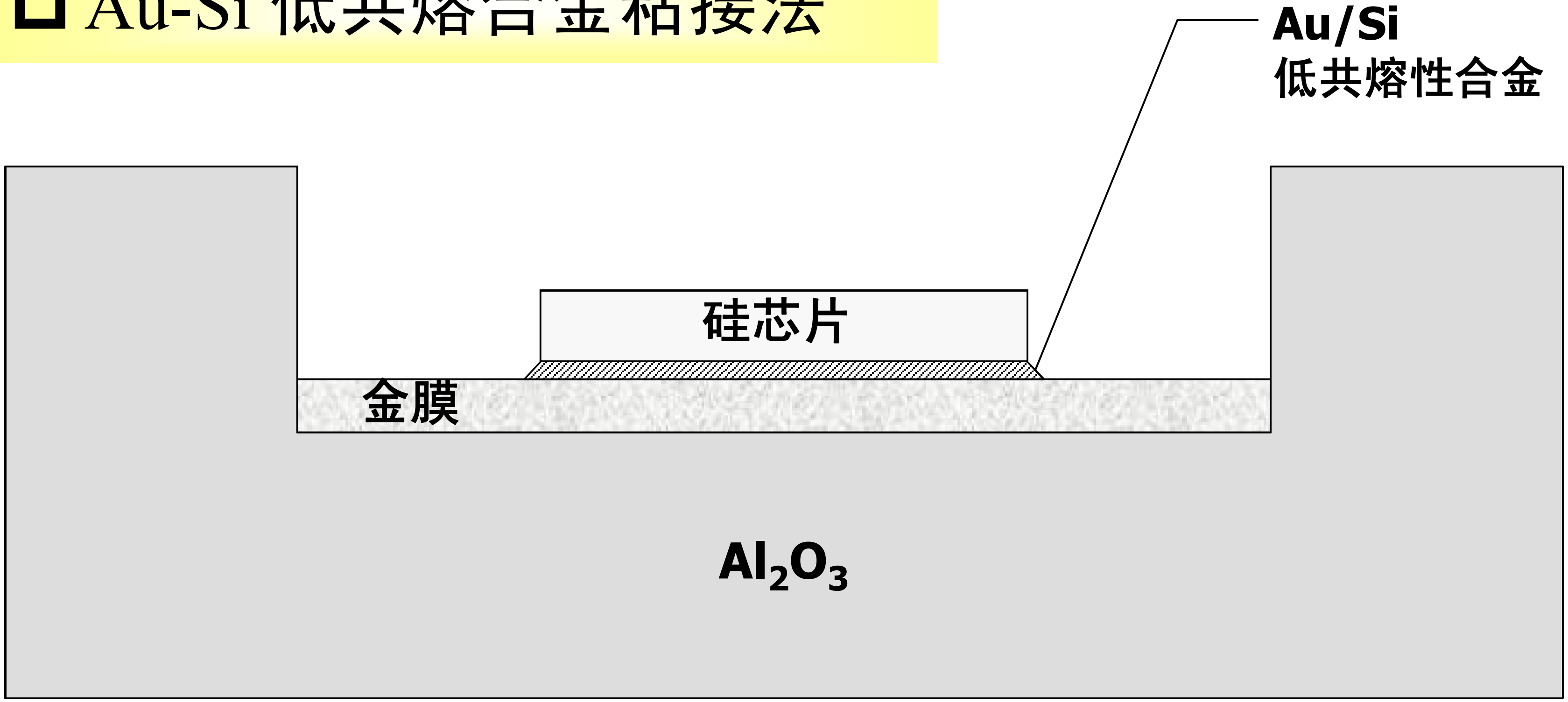
Wire Bonding Technology
-- Die Attach Process

Bonding Type : 环氧树脂 Epoxy



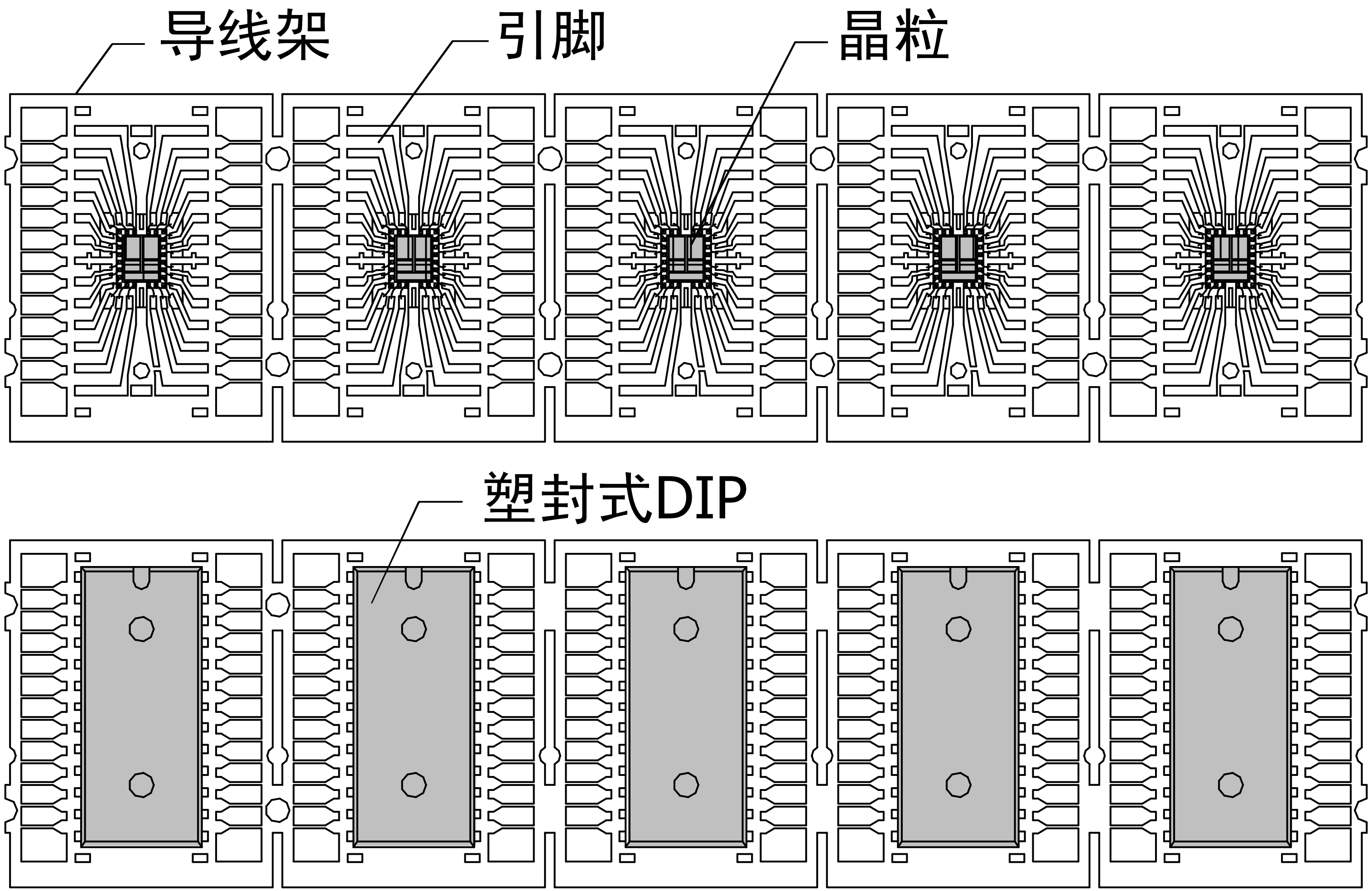
Wire Bonding Technology
-- Die Attach Process

□ Au-Si 低共熔合金粘接法

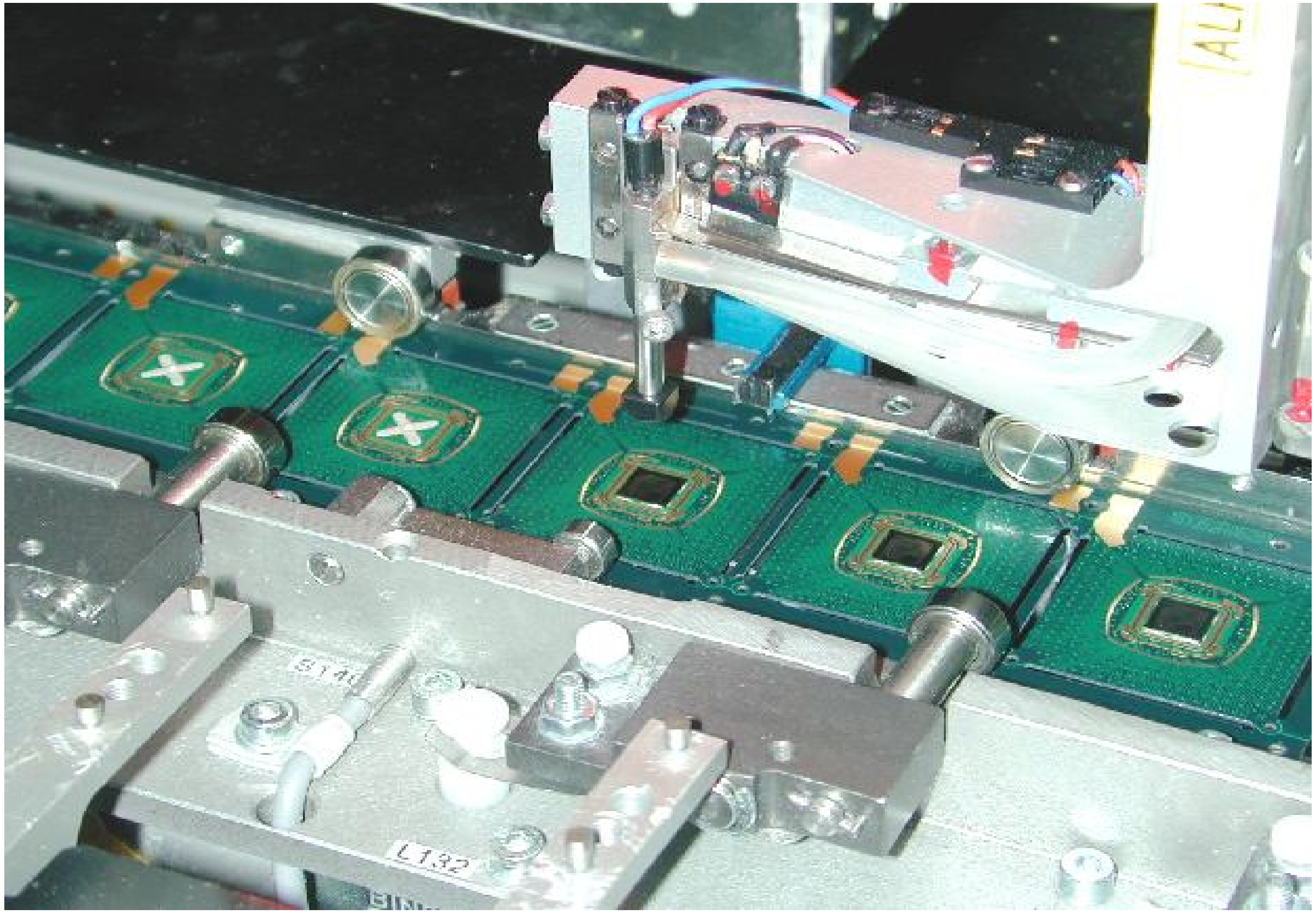


◆ 低共融合金粘接法主要用在芯片产品**需要非常低的背部接触电阻**。

Wire Bonding Technology
-- Die Attach Process

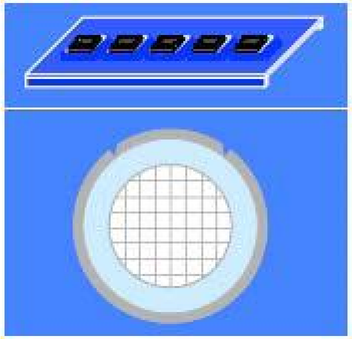


Wire Bonding Technology
-- Die Attach Process



Wire Bonding Technology
-- Die Attach Process

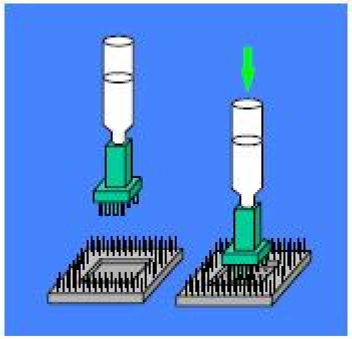
1. Units and Dice/ wafer Load



Objective:

To load the carriers with the units placed on them. To load the dice/wafer into the machine.

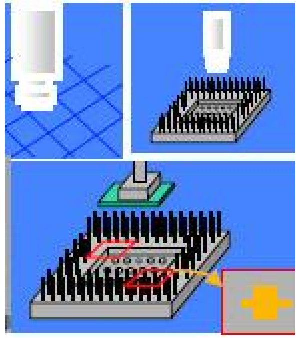
2. Bonding Medium Dispense



Objective:

To dispense the bonding medium onto the substrate die attach paddle.

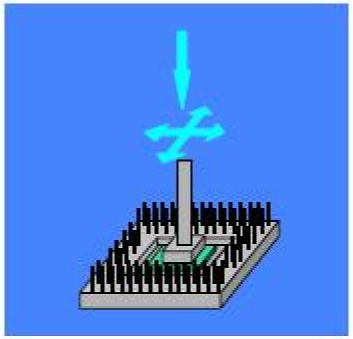
3. Pattern Recognition System (PRS) & Align



Objective:

To align the theta (rotation) position of the wafer. To align the die (X-Y) with respect to the package PRS eye points.

4. Die Attach

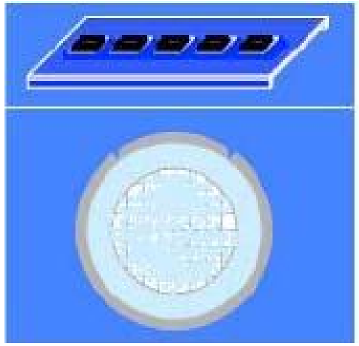


Objective:

To attach the die precisely and form a good adhesion with desired bond line thickness (BLT).

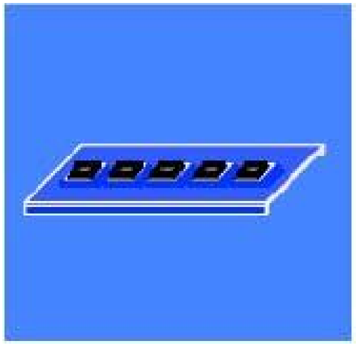
Wire Bonding Technology
-- Die Attach Process

5. Unload (Die Attach)



Objective:
To unload the carriers after die attach. To unload the wafer frame when all good dice are picked up.

6. Load (Cure)

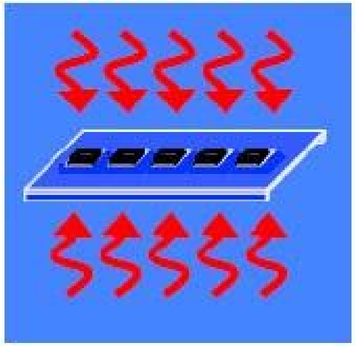


Objective:

To load the carriers

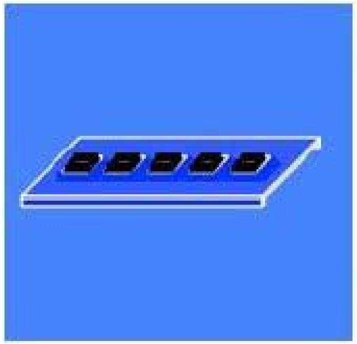
into cure oven.

7. Cure



Objective:
To cure the die attach material to the desirable mechanical, thermal and electrical properties.

8. Unload (Cure)



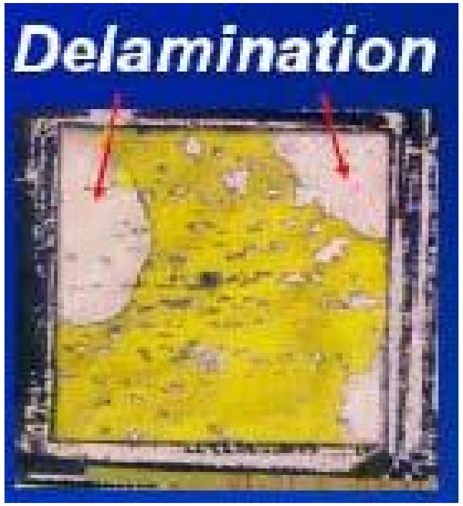
Objective:
To unload the carriers
from cure oven.

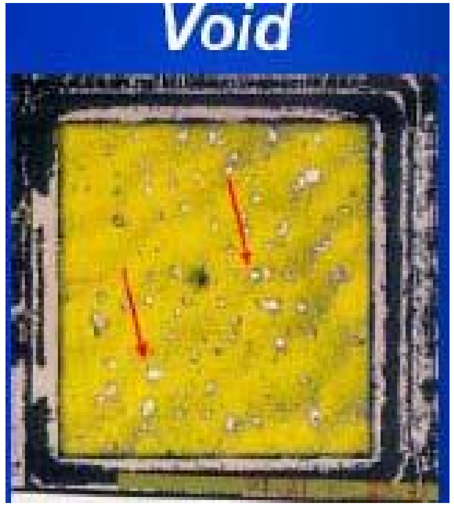
◆ 固化操作仅在adhesive die attach环节使用.

Wire Bonding Technology Die Attach Process Issues and Challenges

□ Issues:

- Ease of process
  - –Delamination control
  - –Void control
- Yield
  - –Adhesive on die
  - –Incomplete wet out/fillet
  - –Die crack
  - –Die placement
- Equipment stability and capability



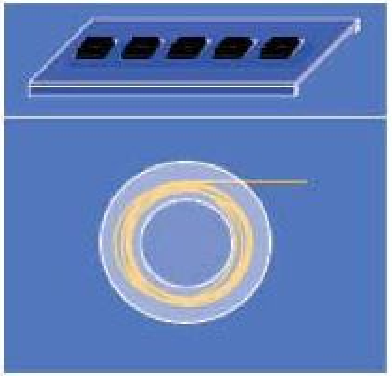


□ Challenges:

- Market requirements drive for very thin die (<3 mils).
- Material selection (e.g. lead free compatible, thermal Material selection and electrical requirements).

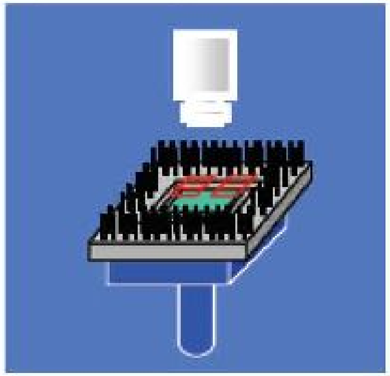
Wire Bonding Technology
-- Wire Bonding Process

1. Units and Wire Load



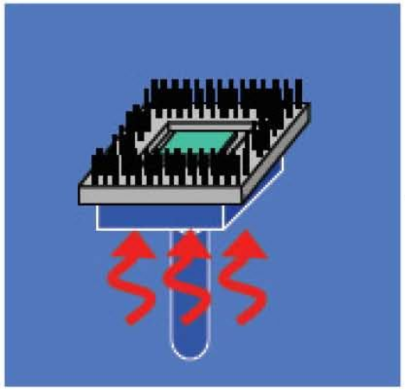
Objective:
To load the carriers with the die attached units placed on them.To load the wire spool into the machine.

3. Pattern Recognition System (PRS) & Align



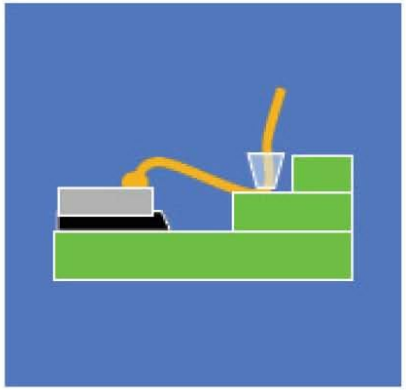
Objective:
To recognize the alignment of the die and the package with their PRS eye points.

2. Unit Preheat



Objective:
To preheat the units for wire bonding (ball bond only).

4. Wire Bond



Objective:
To weld wires onto the die and the substrate pads.

Wire Bonding Technology
-- Wire Bonding Process

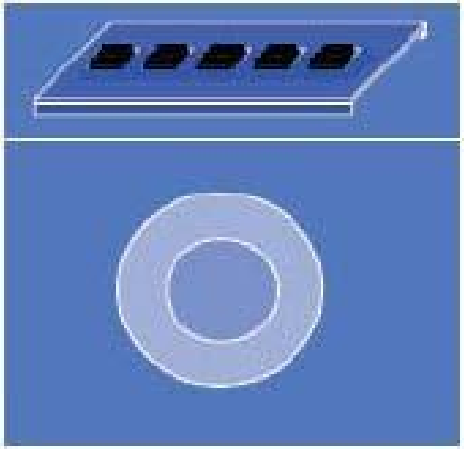
The Wire Bonding Process

5. Unload

Objective:

To unload the carriers after wire bond.

To unload the wire spool when the wire is used up



Wire Bonding Issues and Challenges

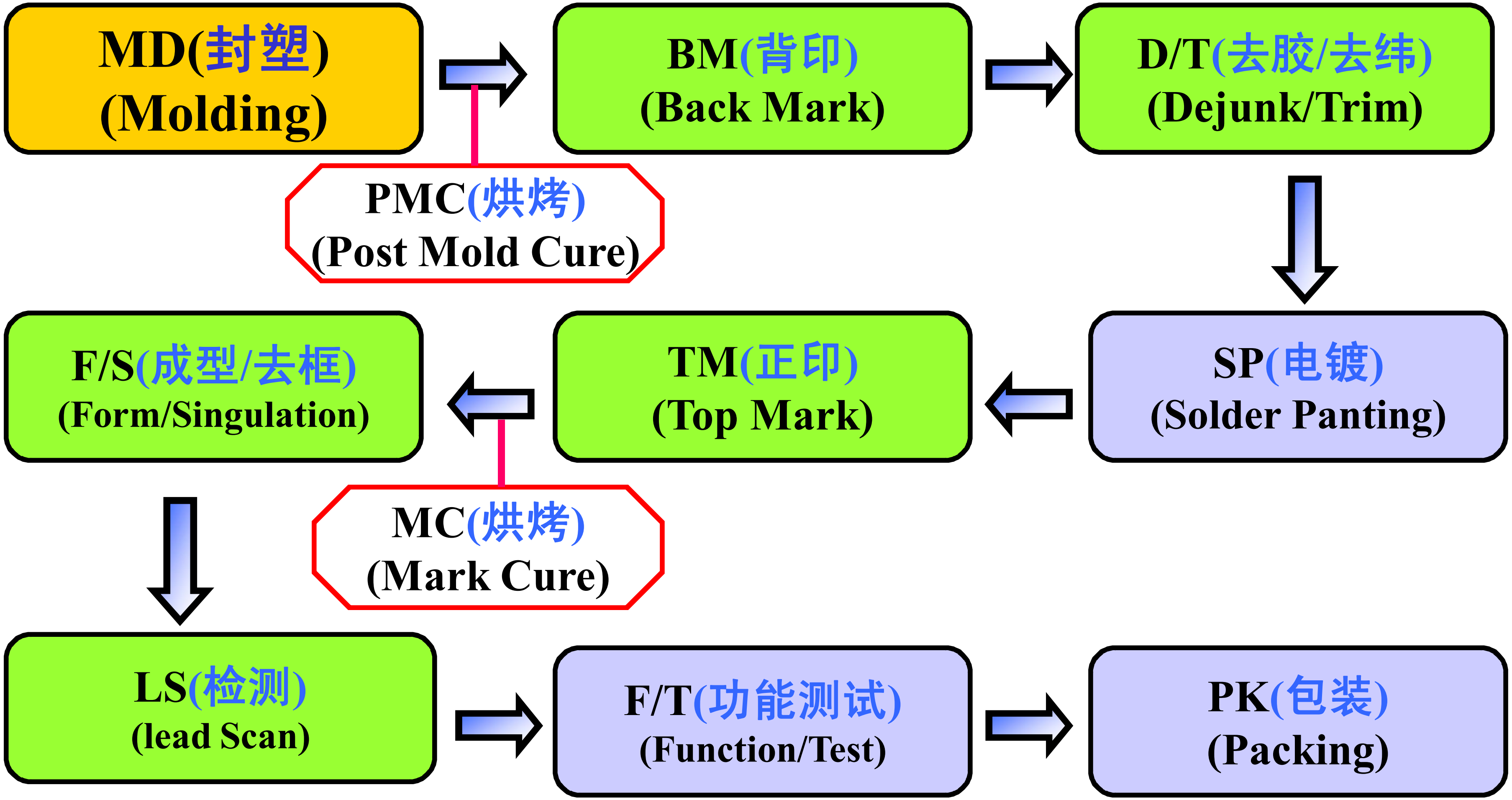
◆ **Issues:**

- **Ease of process**
  - -Looping profile control.
  - -Process optimization for bond ability and bond reliability.
- **Yield**
  - -Lifted bond (non stick on pad or lead).
  - -Sagging and swayed wire. 引线塌陷或歪斜
  - -Tight loop.
- **Equipment stability and capability.**

◆ **Challenges:**

- Market requirements drive for tighter bond pitch. (<37/75um staggered, <60um non-staggered). 交错排列
- Smaller wire diameter (<1.0mils).
- Brittle Intermetallic composition (IMC) on lead free. 无铅工艺中易脆的金属间化合物

IC封装后段流程


MD(封塑)
(Molding)
BM(背印)
(Back Mark)
D/T(去胶/去纬)
(Dejunk/Trim)
PMC(烘烤)
(Post Mold Cure)
F/S(成型/去框)
(Form/Singulation)
TM(正印)
(Top Mark)
SP(电镀)
(Solder Panting)
MC(烘烤)
(Mark Cure)
LS(检测)
(lead Scan)
F/T(功能测试)
(Function/Test)
PK(包装)
(Packing)

Molding

塑封成型

按封装材料分类：

1. 陶瓷封装

(CERAMIC PACKAGE)

2. 塑胶封装

(PLASTIC PACKAGE)

陶瓷封装常用于特殊用途和专业领域IC芯片

例如：高频和军事通讯

◆ **Hermetic lid Sealing**

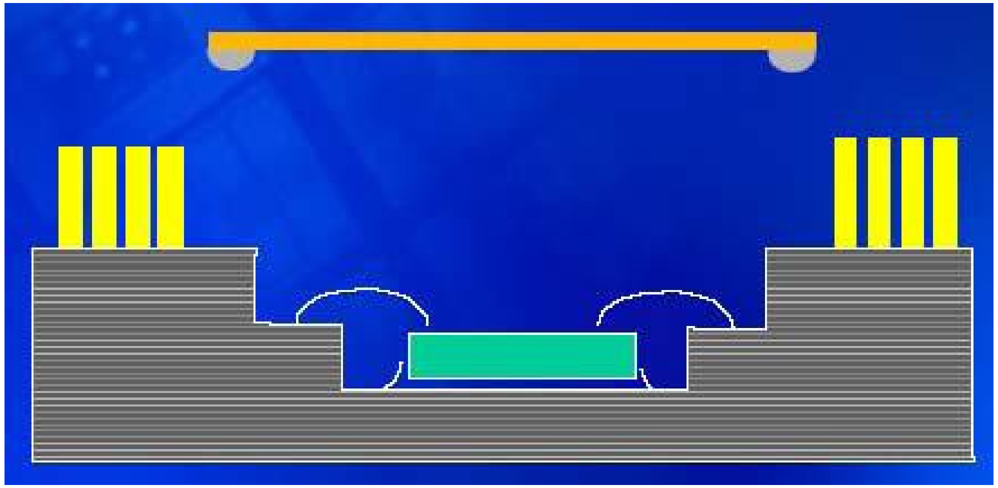
加盖式气密性封装

Hermetic lid Sealing

密封式封装

Method

在第一级互连完成后，将周围印刷有焊料的盖子（或陶瓷，金属或塑料盖）放置在封装基板腔体上（芯片已键合在腔体内）。然后，对整个封装体进行回流，使焊料熔化，完成密封操作。



Sealing Process

Step	Objective
1.Load (lid attach)	To load the wire-bonded units and lids into the lid attach equipment.
2.Pattern Recognition System (PRS)	To recognize the units'X-Y,rotation position by referring to the package PRS eye points for an accurate lid attachment.
3.Lid attachment	To attach the lid over the die cavity with a metal clip.
4.Unload (lid attach)	To unload the carriers after lid attach.
5.Load (reflow)	To load the lid-attached units into the reflow furnace.
6.Reflow	To seal the lid on the substrate.
7. Unload (reflow)	To unload the carriers after reflow.

◆主要分为7个工艺步骤.

Sealing Issues and Challenges

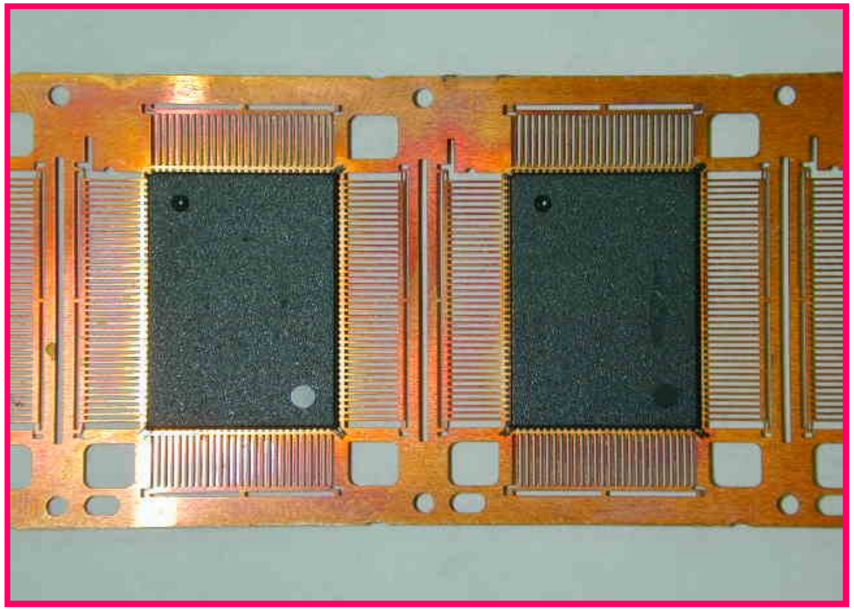
- Issues:
  - Ease of process
    - **Optimization** of lid solder pattern/profile, width and height.
    - **Optimization** of solder reflow profile.
  - Yield
    - Solder **splashes** 锡渣
    - Test short due to **free solder balling** in cavity.
    - Lid misplacement.
  - Equipment stability and capability
- Challenges:

  - **Lead free** compatible solder is expected.

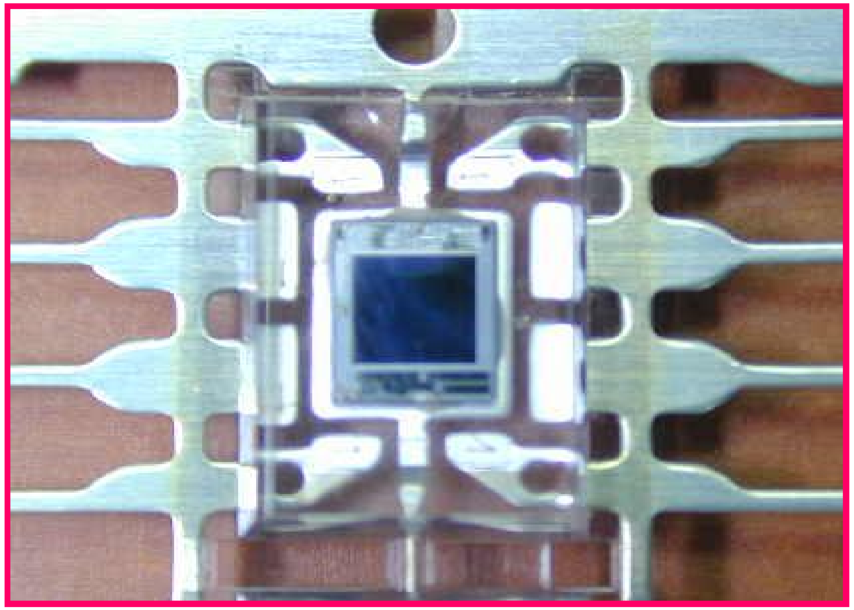
2. 塑料封装 (PLASTIC PACKAGE)

适于大规模生产、为目前主流封装形式。

(市场占有率约为90%)



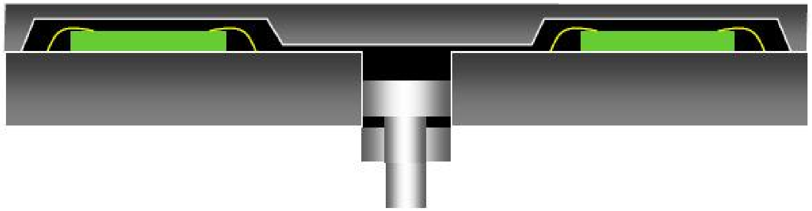
黑胶-IC



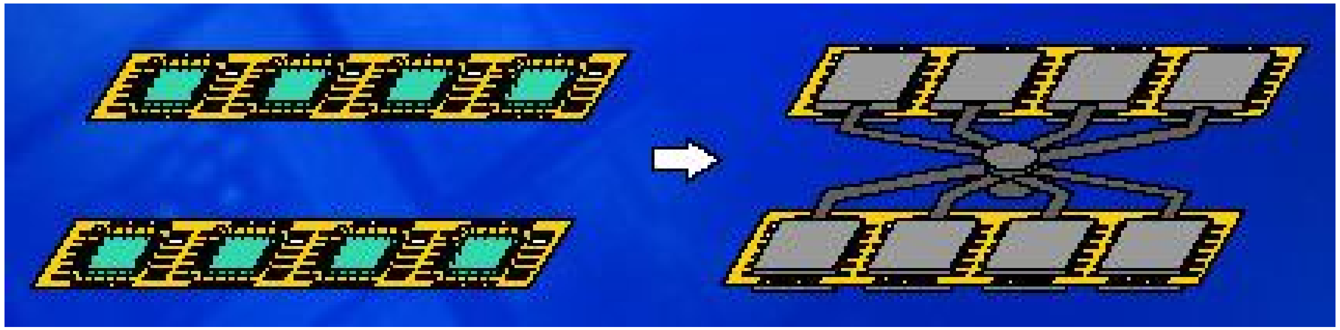
透明胶-IC

2. 塑料封装 (PLASTIC PACKAGE)

□ Molding Method



- ✓ 成型是用封装材料将芯片和一级互连完全包裹住，用来阻止外部压力或环境对芯片带来损伤。
- ✓ 封装体通过个体或批处理的方式进行高温退火，实现封装材料所需的机械应力和防潮等特性。



Material

材 料

Epoxy Molding Compound

- IC塑料封装材料为热固型环氧树脂 (EMC)，其作用为填充模穴(Cavity)将导线架(L/F)完全包覆，保护好已焊接引线的芯片。

Lead Frame

- Lead Frame即导线架，其目的是承载晶片以及焊接金线用，使信号得以顺利传递，而达到系统的需求。

◻ Mold Compound (MC) 塑封化合物材料

What is mold compound (MC) ?

It is a solid epoxy/phenolic based resin with fillers

固态环氧树脂或酚醛树脂填充物

Function:

Provides encapsulation to protect the wire and die

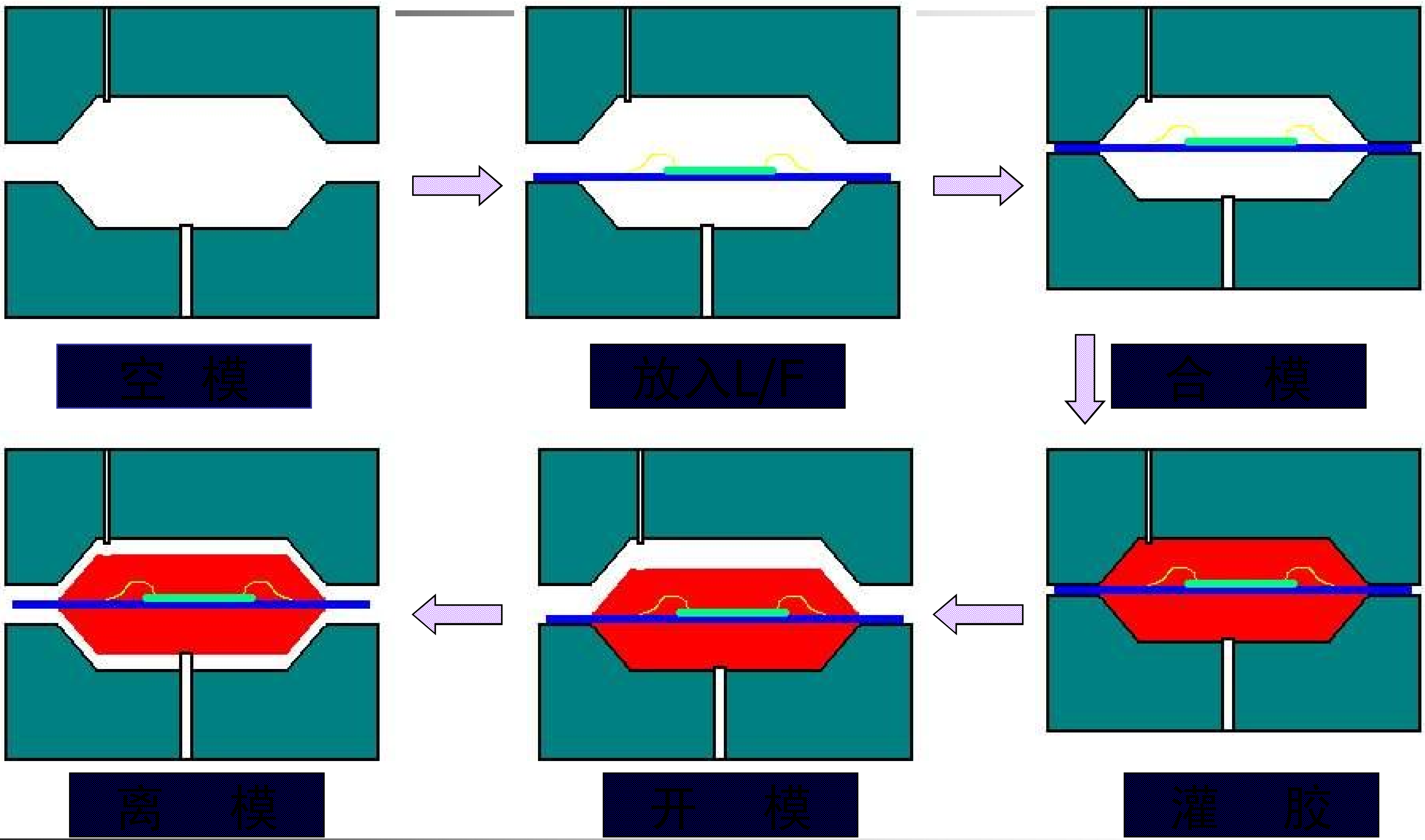
Challenges, trends and potential areas of research:

Pb free compatible MC- able to withstand high temperature reflow

Process simplification – elimination of post mold cure and no plasma clean

Flow modeling – simulation of mold flow to eliminate voids

MOLDING CYCLE



Mold Process

Step	Objective
1. Load (Prebake)	To load the units into the prebake oven (usually manually).
2. Prebake	To remove residual moisture from the die and the package that will react with the coupling agent in the mold compound, causing random voids in the mold. Load/unload of the units (in specially designed cassettes) are usually done manually.
3. Unload (Prebake)	To unload the units from the prebake oven (usually manually).
4. Load (Plasma clean)	To load the units (in specially designed cassettes) into the prebake oven (usually manually).

◆ It is broken down into 16 major steps

Mold Process (cont.)

5. Plasma clean	To clean the exposed surfaces of the die and the package in order to improve the adhesion of the mold compound. Load/unload of the units (in specially designed cassettes) are usually done manually.
6. Unload (Plasma clean)	To load the units (in specially designed cassettes) into the plasma chamber (usually manually).
7. Load (Mold)	To load (manually or automatically) the units onto the lower mold half.
8. Positioning and clamping	To position the units onto the registration pins on the mold. The mold halves are then brought together into the clamped position.
9. Mold compound preparation	To place the thermoset molding compound into the transfer pot of the molding tool for preheating.
10. Strip preheating	To preheat the strip to ease molding.

Mold Process (cont.)

11. Mold transfer, cure and unit separation	To transfer the molding compound into the mold runner system (by the plunger).To cure the mold compound at a preset time. To separate the individual units after cure (cull breaking).
12. Unload	To unload the units after molding.
13.Load (PMC)	To load the units into the cure oven (usually manually).
14. Post mold cure	To completely cure the mold compound for the desirable mechanical properties.
15.Unload (PMC)	To unload the units from the cure oven (usually manually).
16.Deflashing/ Dejunking (if needed)	To remove the excess molding compound or the thin flash of molding compound (known as resin bleed or flash) from the strips. (i. Media deflashers, ii. solvent deflashers and iii. water deflashers)

Mold Issues and Challenges

Issues

- **Ease of process**
  - **–Wire sway due to the flow of molding compound.**
  - **–Post mold warping of strip or unit due to the flow-induced stress.**
- **Yield**
  - **–Excess mold flash.**
  - **–Incomplete molding.**
  - **–Voids.**
- **Equipment stability and capability**

Challenges

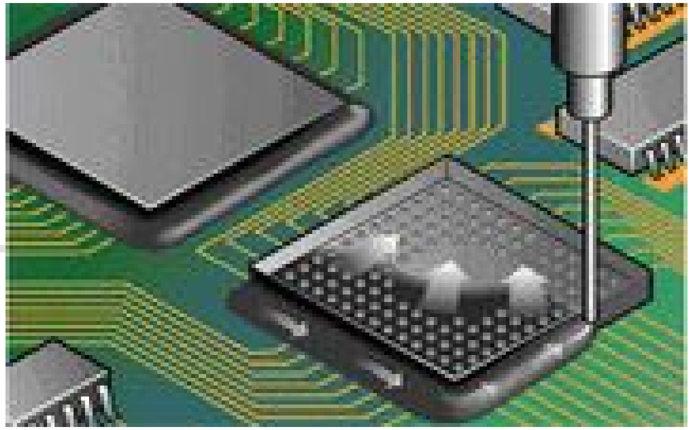
⇒ Tighter bond pitch (<37/75um staggered, <60um non-staggered) requires

⇒ Smaller filler size in molding compound.

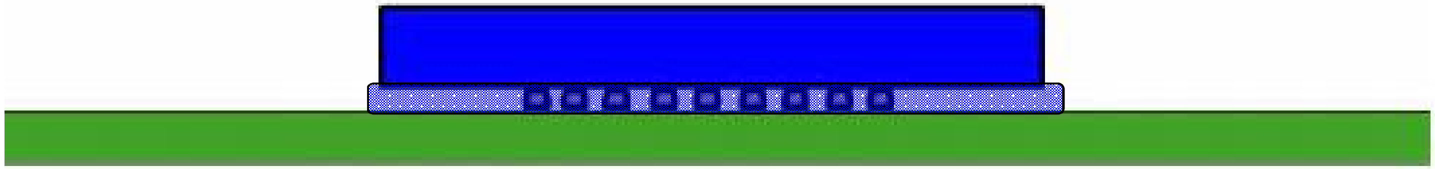
Smaller wire diameter (<1.0mils) which is more prone to wire sway.

⇒ Lead free compatible molding compound material.

Flip Chip Underfill



□ Method

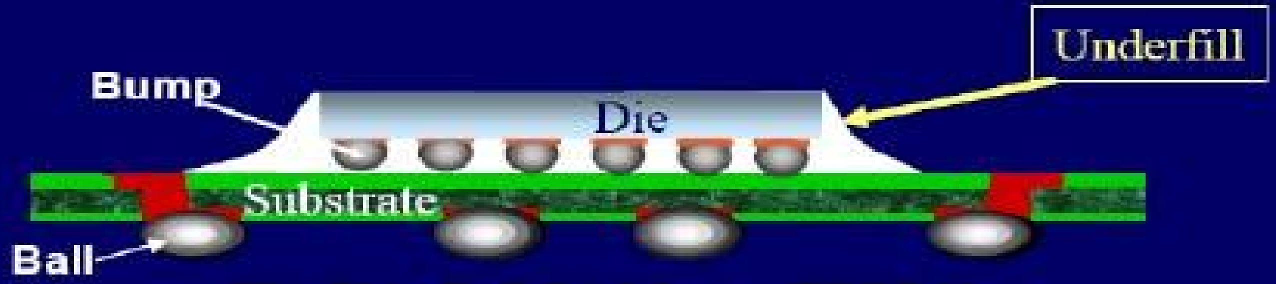


倒装芯片第一级互连环节完成后，封装体开始第一次**干/湿预烘烤**，**消除基板湿气**。

接着，对整个封装体**进行预加热**至预设温度后，开始利用线性针嘴**在芯片底部填充环氧树脂**，使得填充材料覆盖整个模具和第一级互连。

然后，对封装体**进行高温退火**，实现封装材料所需的**机械应力和防潮性能要求**。

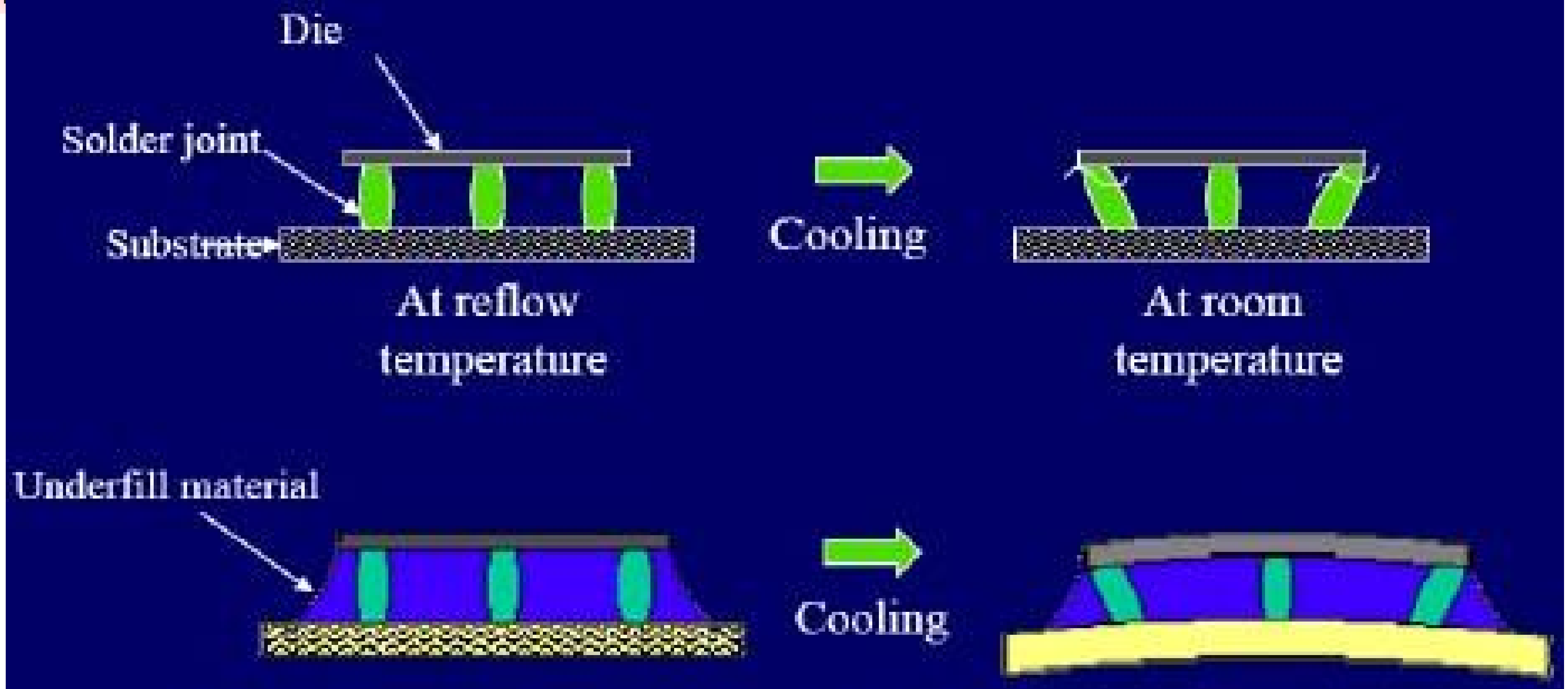
Why Underfill?



Flip Chip Ball Grid Array (FCBGA)

- In flip-chip technology, the gap between substrate and chip is underfilled with highly filled epoxy system
  - High modulus, low CTE adhesive which couples the die and substrate
- Role of underfill:
  - Provides reliability to the flip chip package
    - By redistributing the stress due to CTE mismatch
    - Prevents interconnect fatigue by applying compressive stresses to the bumps

Mechanism of UF Encapsulation

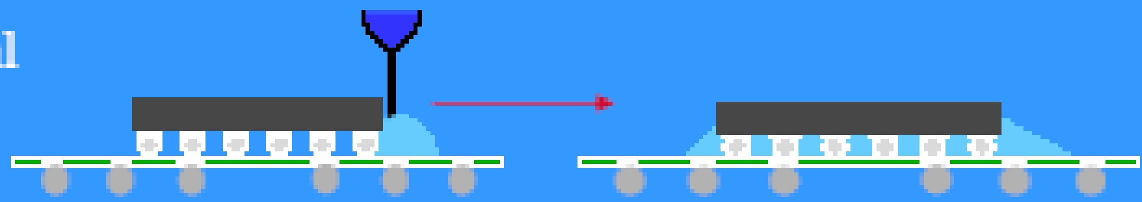


Substrate and chip are interlocked by underfill
The strain on joint is converted to deformation
Joint is compressed and protected by underfill

Underfill Technology Options

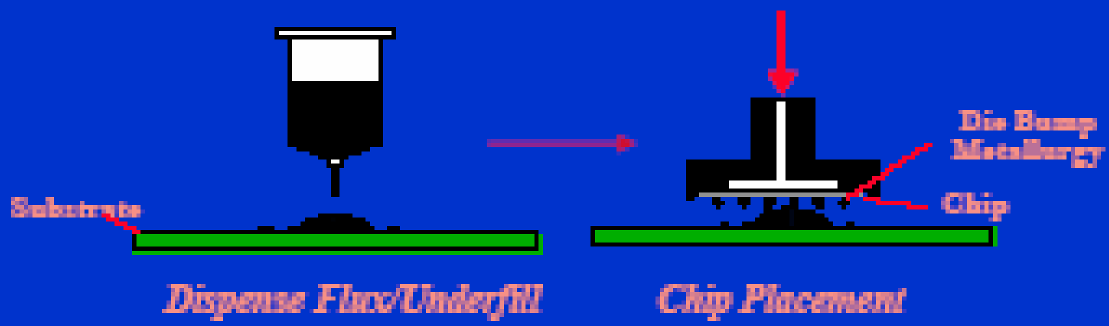
Capillary Underfill (CUF)

Flow of underfill material underneath die is due to capillary action.



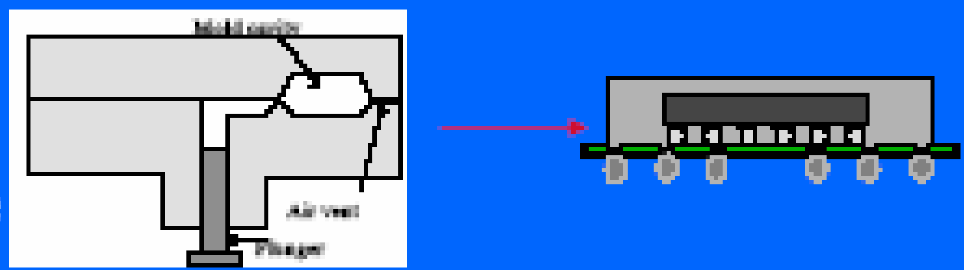
No flow Underfill (NUF)

Die attach concept of underfilling using one step curing and reflow



Mold Underfill (MUF)

Transfer of molten mold compound to encapsulate the die and fill the gap between substrate & die.



Underfill Process

It is broken down into 10 major steps

Step	Objective
1. Load (Underfill)	To load the chip-attached units into the prebake oven. To load the underfill syringe into the dispenser.
2. Prebake*	To remove the moisture absorbed in the package that will cause random voids in the underfill.
3. Preheat	To preheat the parts exited from the prebake oven to a desired temperature to ease the flow of underfill material later.
4. Pattern Recognition System (PRS)	To recognize the units' X-Y, rotation position by referring to the package PRS eye points for an accurate dispensing.
5. Dispense material	To dispense the underfill material at the side of the die.

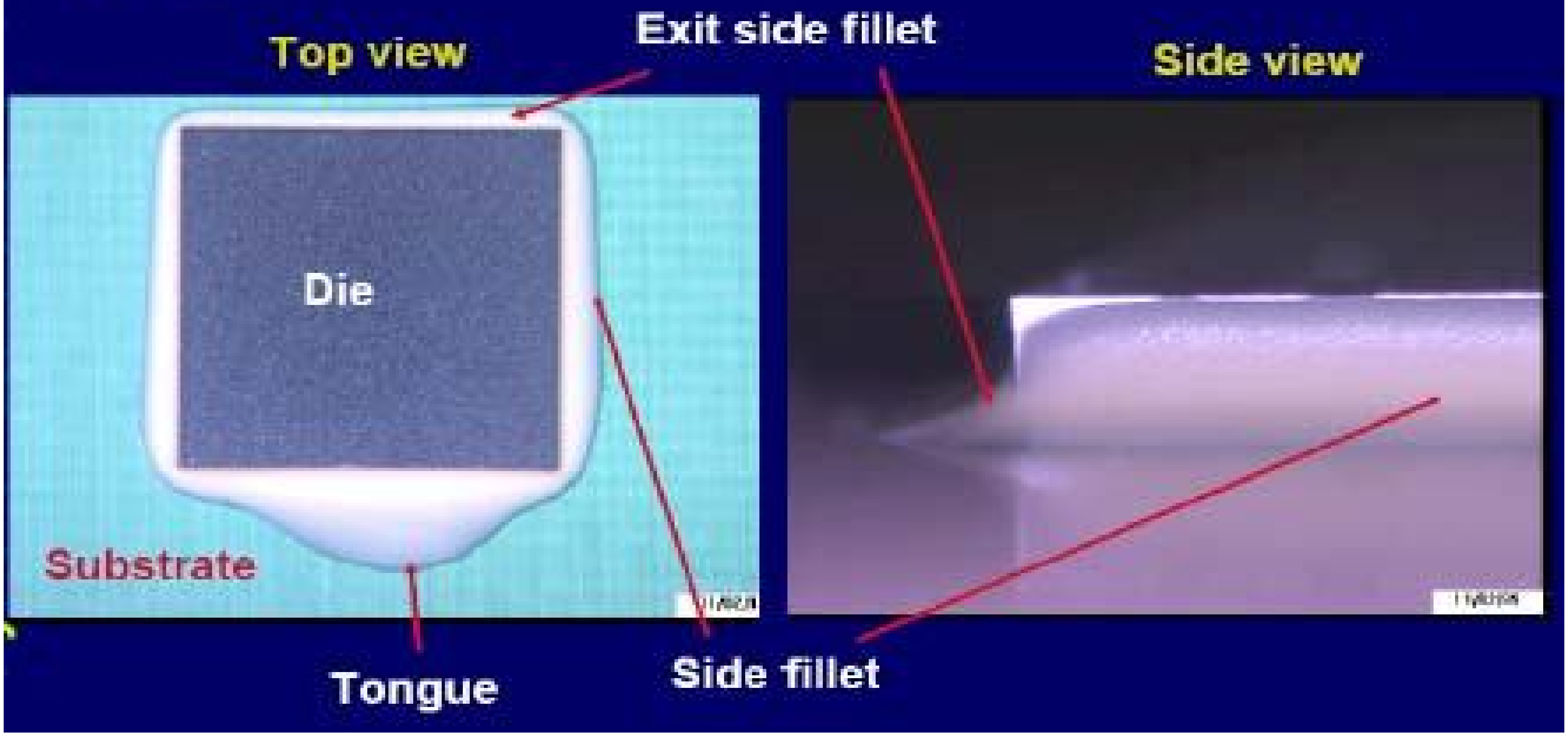
Underfill Process (cont.)

6.Post-dispense heating	To promote the capillary flow the underfill material to cover the entire die area.
7. Unload(Underfill)	To unload the carriers after underfill.
8.Load (Cure)*	To load the carriers into the cure oven.
9.Cure	To cure the encapsulation material to the desirable mechanical properties.
10. Unload (Cure)	To unload the carriers from the cure oven.

- Underfill is more sensitive to moisture and can form voids which affect the part reliability.
- Prebake and dispense are usually linked to avoid extended exposure for moisture absorption again after prebake.

Self-filleting Property of CUF

- Fillets and tongue are formed at the sides of the die. The tongue is formed at the dispense side and the other sides are fillets



Underfill Issues and Challenges

Issues

- Ease of process
  - –Optimization of dispense weight and pattern.
  - –Fillet width control especially at the dispense side (s).
  - –Underfill voiding control.
- Yield
  - –Underfill material on die.
  - –Incomplete filleting at opposite dispense side(s).
- Equipment stability and capability

Challenges

⇒ Lead free compatible underfill material is expected.

⇒ Tighter bump pitch requires smaller filler size in underfill material.

⇒ Tighter bump pitch is more prone to void formation.

小间距引脚更容易造成虚空。

Marking

Purpose

在封装体表面，刻写清晰易读的产品**标识数据**。

标识数据包括有**编码、生产日期、生产厂家**和**原产国**等。

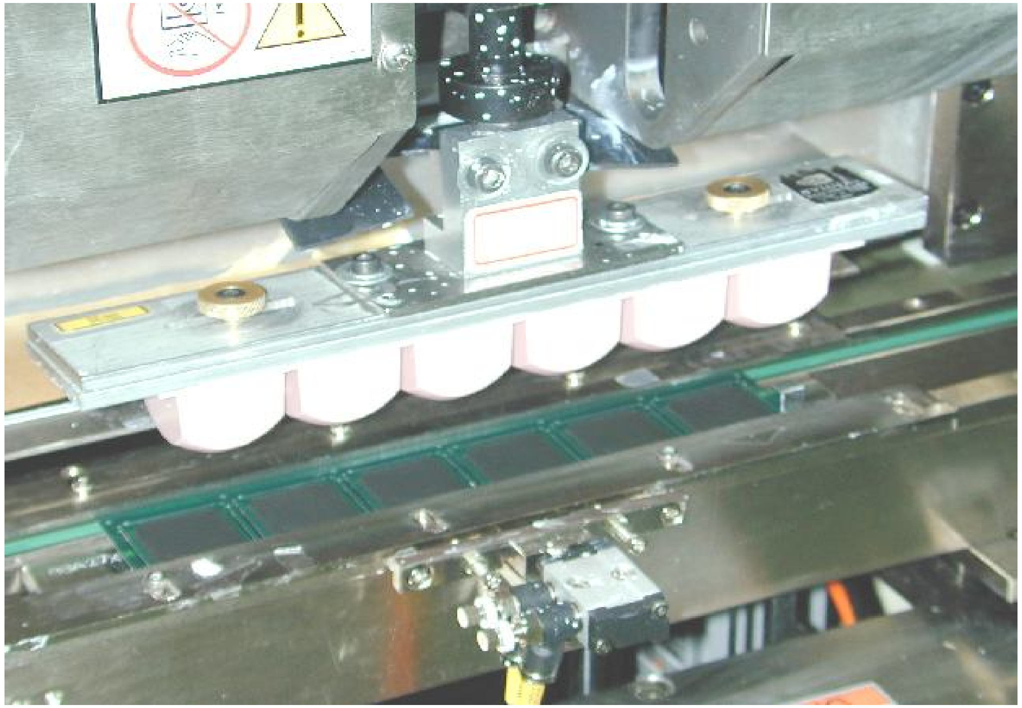
Marking Process Methods

1) Laser marking /writing 激光打标
2) Polymer-based ink jetting 聚合物喷涂
3) Polymer-based ink stamping. 聚合物模压

Method

- For **laser marking**, trays of units are indexed under the laser, which burns the product name and **bin information** onto the unit (die or substrate surface).
- Similar process is used for **ink jetting** with the polymer-based ink except that ink is sprayed onto the unit surface.
- For **ink stamping** process, a stamp is used to transfer ink from a ink pad to the unit surface.

Top/Bottom Ink / Laser Marking

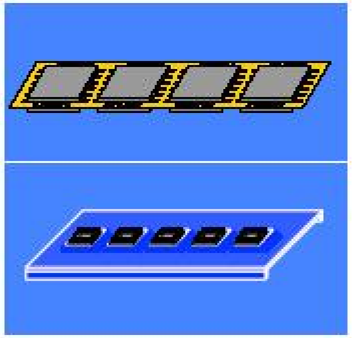


- **Method:** Ink mark / Laser mark
- **Key:** H_2 flame to remove wax(蜡) on molding compound
- **Check:** Marking permanency

Marking Process

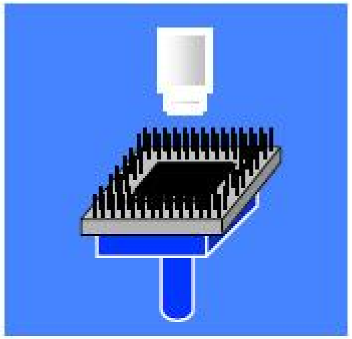
- The MARKING process is broken down into 4 major steps:

1.Units Load



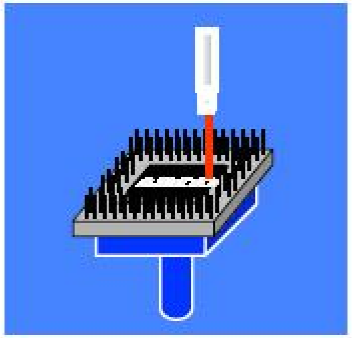
Objective:
To load the molded strips or carriers with the encapsulated units placed on them.

2.Pattern Recognition System (PRS)



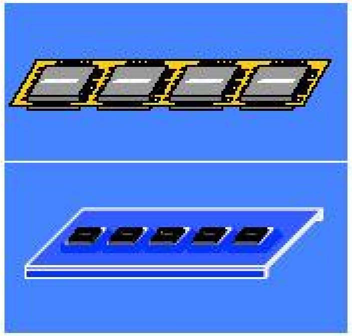
Objective:
To recognize the position of the encapsulated units or strips with their PRS eye points.

3.Marking



Objective:
To mark the encapsulated units or strips.

4.Units Unload



Objective:
To unload the marked units or strips.

Marking Issues and Challenges

☐ Issues

- **Ease of process**
  – Optimization of the laser mark depth in avoiding reliability impacts to the marked surface.
- **Yield**
  – Legibility issue. 易辨认
  – Offset marking. 偏移
  – Ink smearing. 污点、拖尾效应
- **Equipment stability and capability**

☐ Challenges

- Smaller form factor with limited marking area.
- Laser marking on the glob top encapsulation surface can initiate cracks.

Second level interconnection

☐ Purpose

The second level interconnection process is to prepare the interconnection structure on the units to form joints with the board.

☐ Process Methods

1、 Lead fingers

☐ by trim-and-form

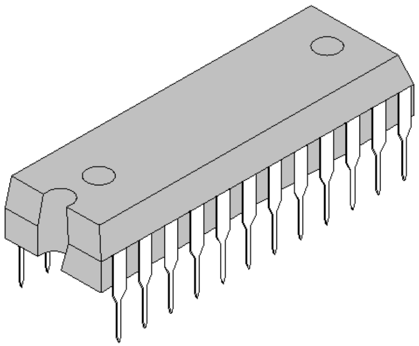
2、 Balls attach

☐ by gravity or vacuum; flux or solder paste

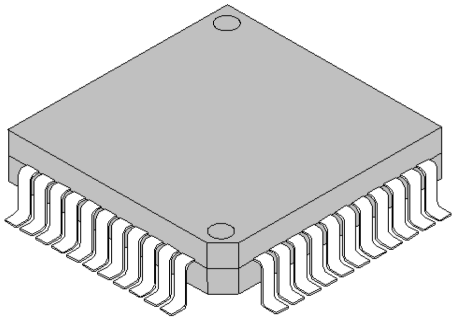
1、 Lead fingers

Dejunk/Trim and Singulation/Form

(去胶/去纬 & 去框/成 型)



双排引脚封装
(DIP)



四方扁平封裝
(QFP)

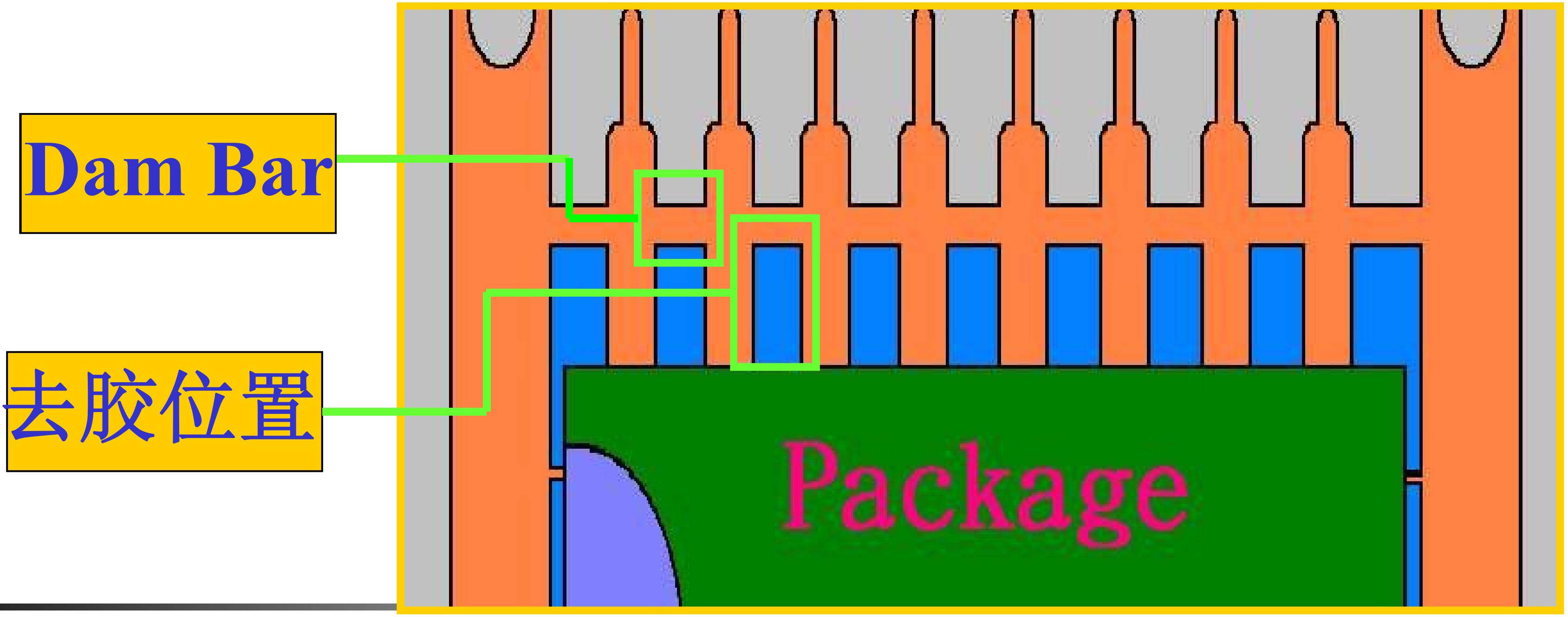
1、 Lead fingers

□ Trim and Form Method

- 去纬成形是通过专用工具 **trim-and-form tool**完成。
- 首先把连接所有引脚的金属框架剪断，接着将所有引脚形成预期的形状。
- 再将引脚涂覆焊料。涂覆焊料通过焊料浸渍（后切割）或焊料电镀（前切割）等办法实现。

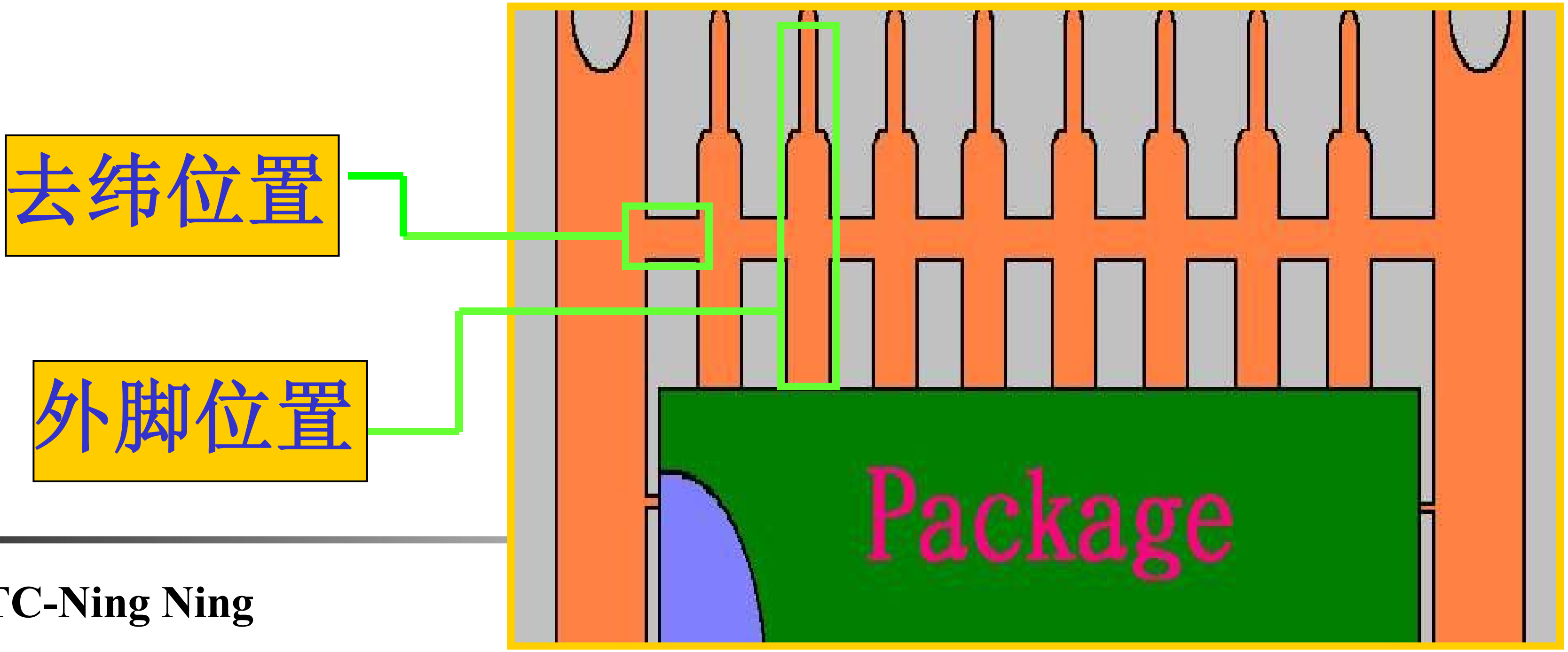
(1) 去胶（Dejunk）

目的: 指利用机械模具将引脚间的废胶去除；亦即利用冲压的刀具（Punch）去除掉介于胶体（Package）与障碍杆（Dam Bar）之间的多余胶体。

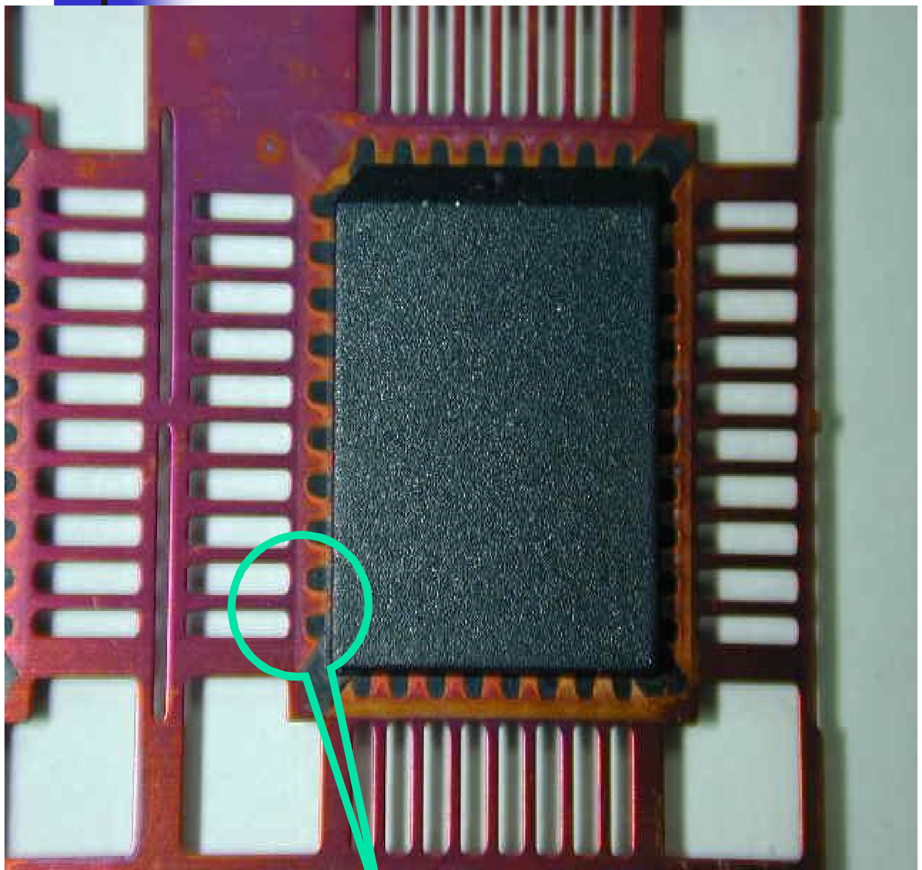


(2) 去纬（Trimming）

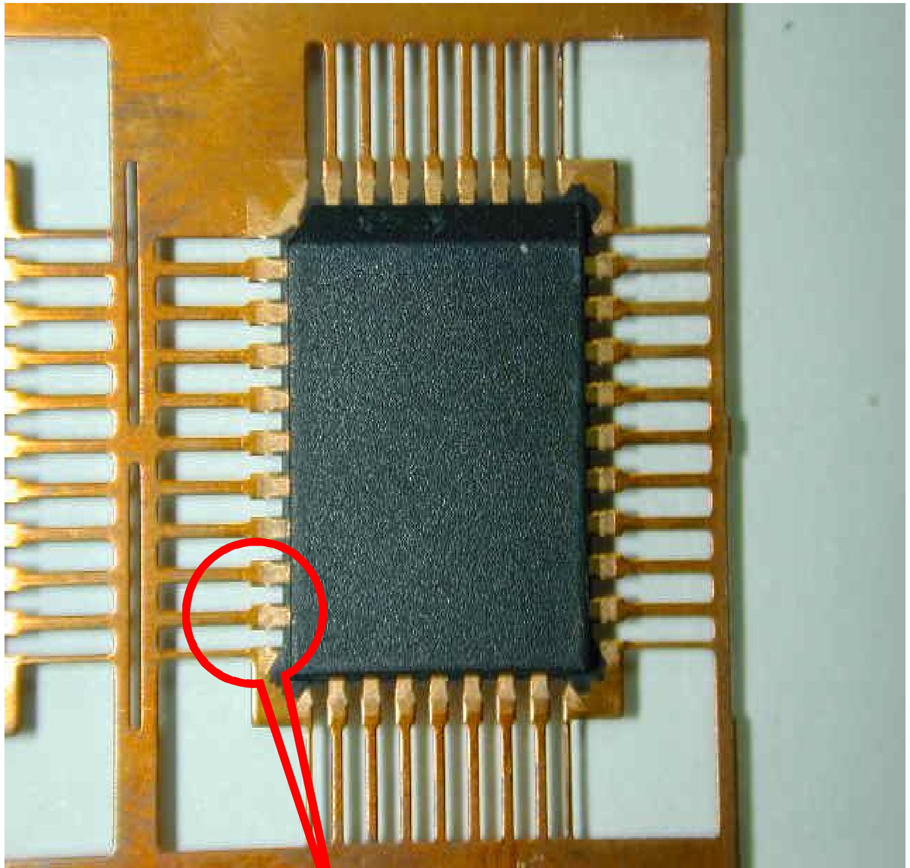
目的： 指利用机械模具将引脚间金属连接杆切除。由于导线架（Lead Frame）内脚（Inner Lead）已被胶体（Compound）固定于Package里，利用Punch将Dam Bar切断，使外脚（Out Lead）与内部线路导通，成为单一通路而非相互连接。



去胶/去纬 (Dejunk / Trimming)



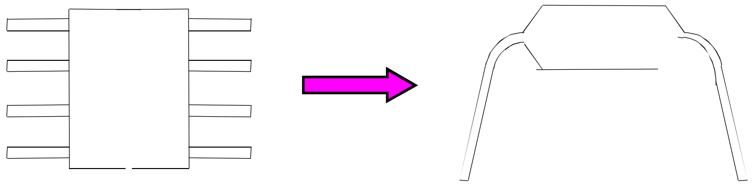
去胶/去纬前

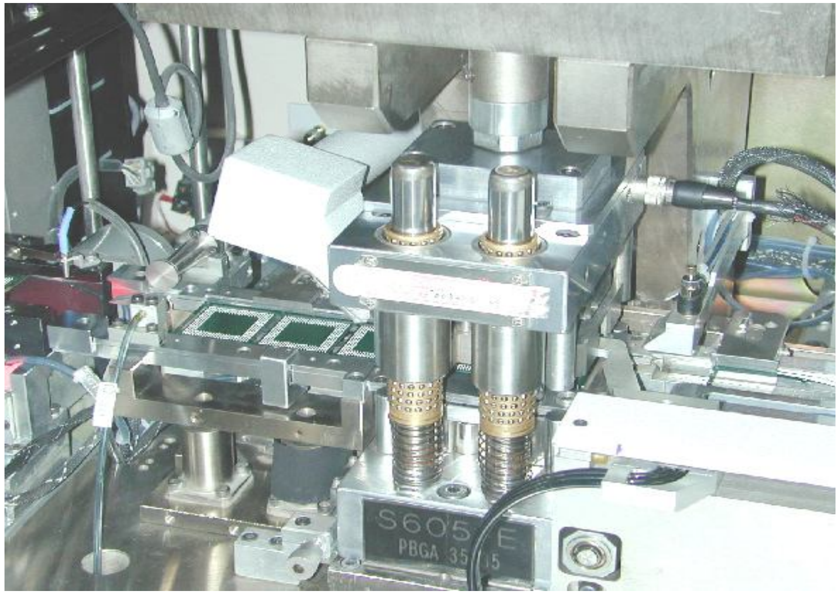


去胶/去纬后

Singulation / Forming

(去框/成 型)





□ 去框剪切的目的

- ◆将整条导线架上已封装好的晶粒，每个独立分开。同时要把不需要的连接用材料及部份凸出树脂切除(**dejunk);**
- ◆剪切完成时的每个独立封胶晶粒的模样，是一块坚固的树脂硬壳并由侧面伸出许多支外引脚。

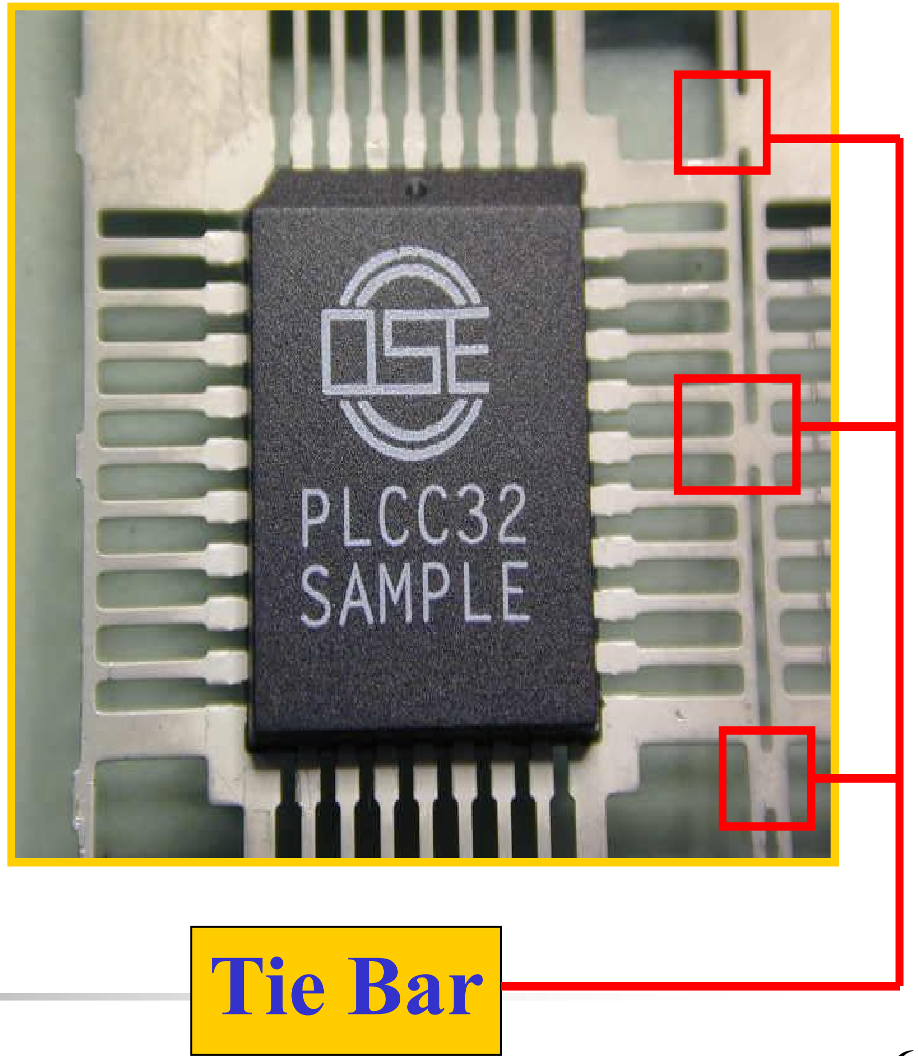
□ 成型的目的

- ◆将这些外引脚压成各种预设的形状，以便于后续装置在电路板上使用；
- ◆由于定位及动作的连续性，剪切及成型可在一部机器上，或分成两部机器（**trim / dejunk , form / singular**）连续完成；
- ◆成型后的每颗ＩＣ送入塑料管（**tube**）或承载盘（**tray**）以便输送。

(3)去框（Singulation）

目的：

将已完成盖印（Mark）制程的Lead Frame，以冲模的方式将连接杆（Tie Bar）切除，使Package与Lead Frame分开，方便下一个制程。



Singulation 去框 (分离/切单颗)

- **Purpose:**

The singulation process is to separate finished units from the strips or tapes or lead frames.

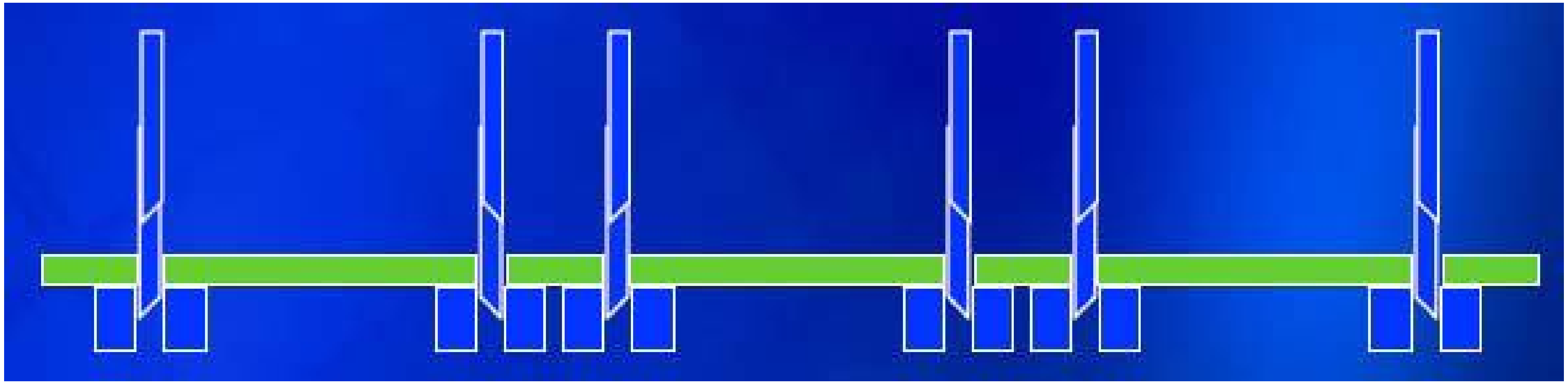
- **Process Methods:**

1) Punching 冲压
2) Sawing / dicing 锯/切
3) Laser cutting (Nd: YAG laser system**)**
4) Bend and slap 弯曲折断
5) Shearing 剪切

Singulation(自动成型机)

Singulation Process Methods

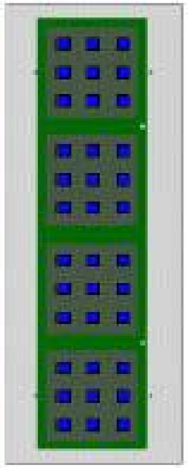
1) Punching method

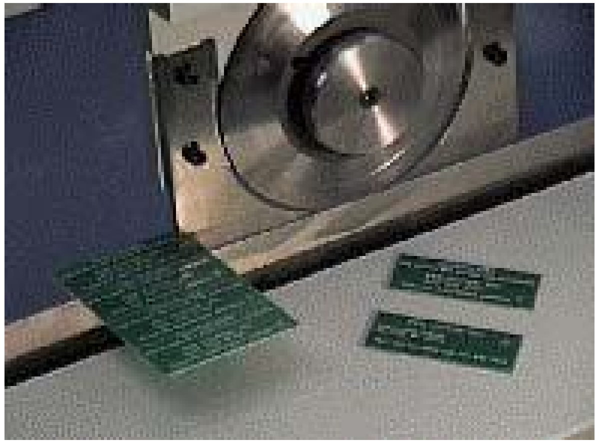


Punch singulation operation is conducted by **mechanical shearing** with the tooling made of high speed **steel material**. A few units can be punched at one time.

Singulation Process Methods

2) Saw Singulation method





The strips are first loaded into a saw station. The saw machine then separates the units with sawing blades. Units will later be cleaned and picked up from the nest to a shipping tray.

Singulation Process Methods

3) Laser cutting method

Units (usually CSP) are singulated with a focused laser beam system while being moved on a high speed* and high precision X-Y translation stage. A high-speed jet of gas is used to blow away the debris and prevent excessive accumulation of heat to the CSP surface during the singulation process. Not commonly used.

Note: * The laser cutting process is able to achieve 70 mm/s for IC packages with 1.1 mm thickness.

Singulation Process Methods

4) Bend and slap method
弯曲折断

Developed to aid in the breaking of edge strips from scored and routed panels. Not commonly used.

5) Shearing method
剪断

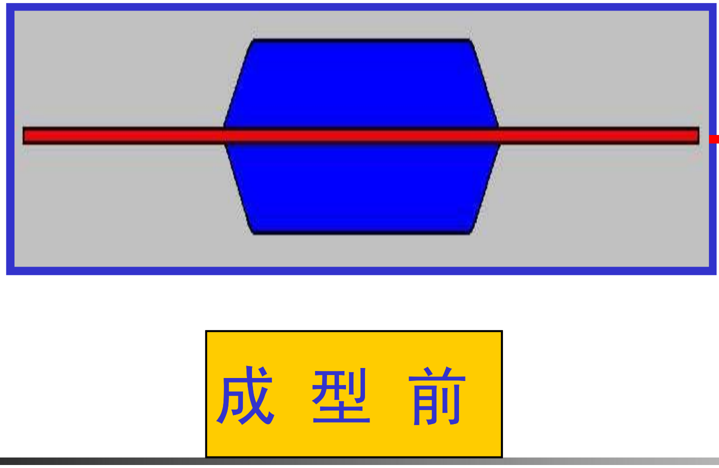
A high stress method to singulate units. Not commercially available.

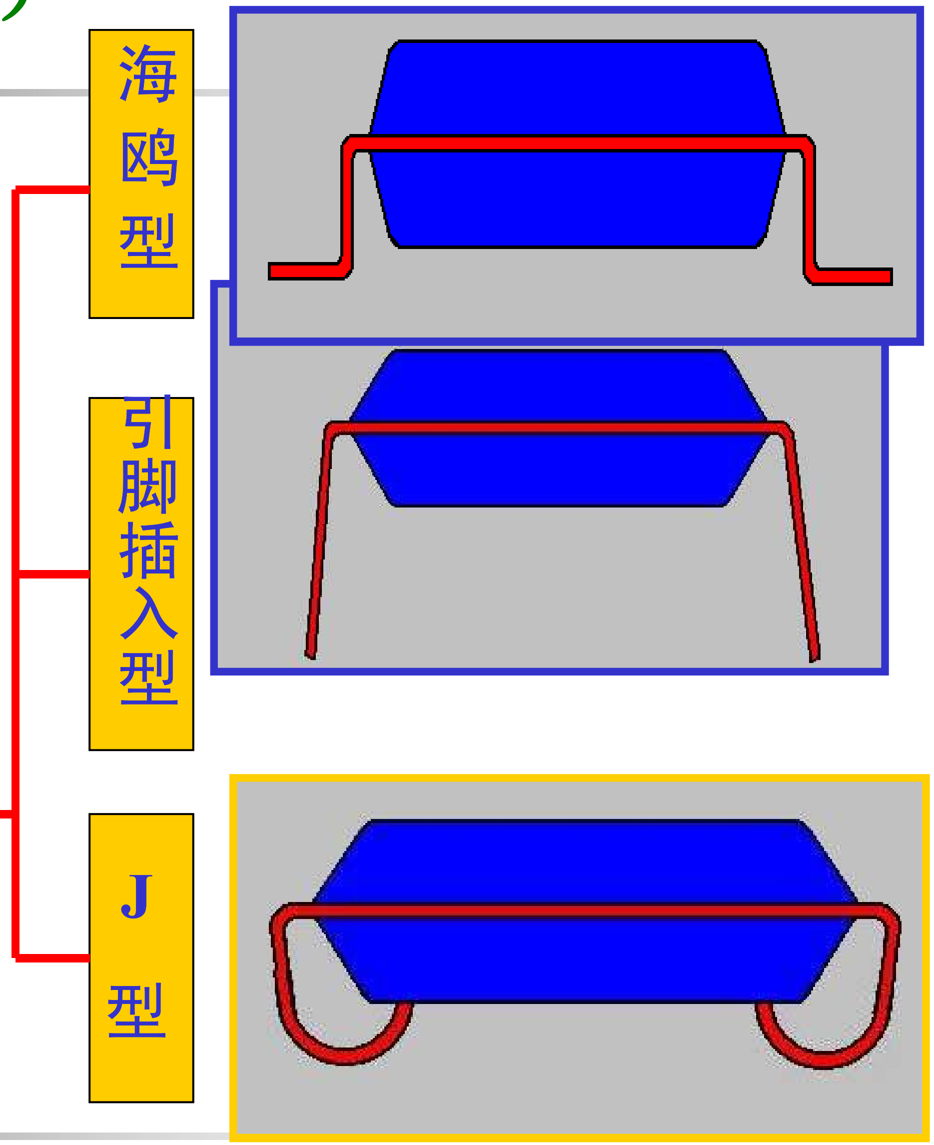
(4)成型（Forming）

F/S 的型式

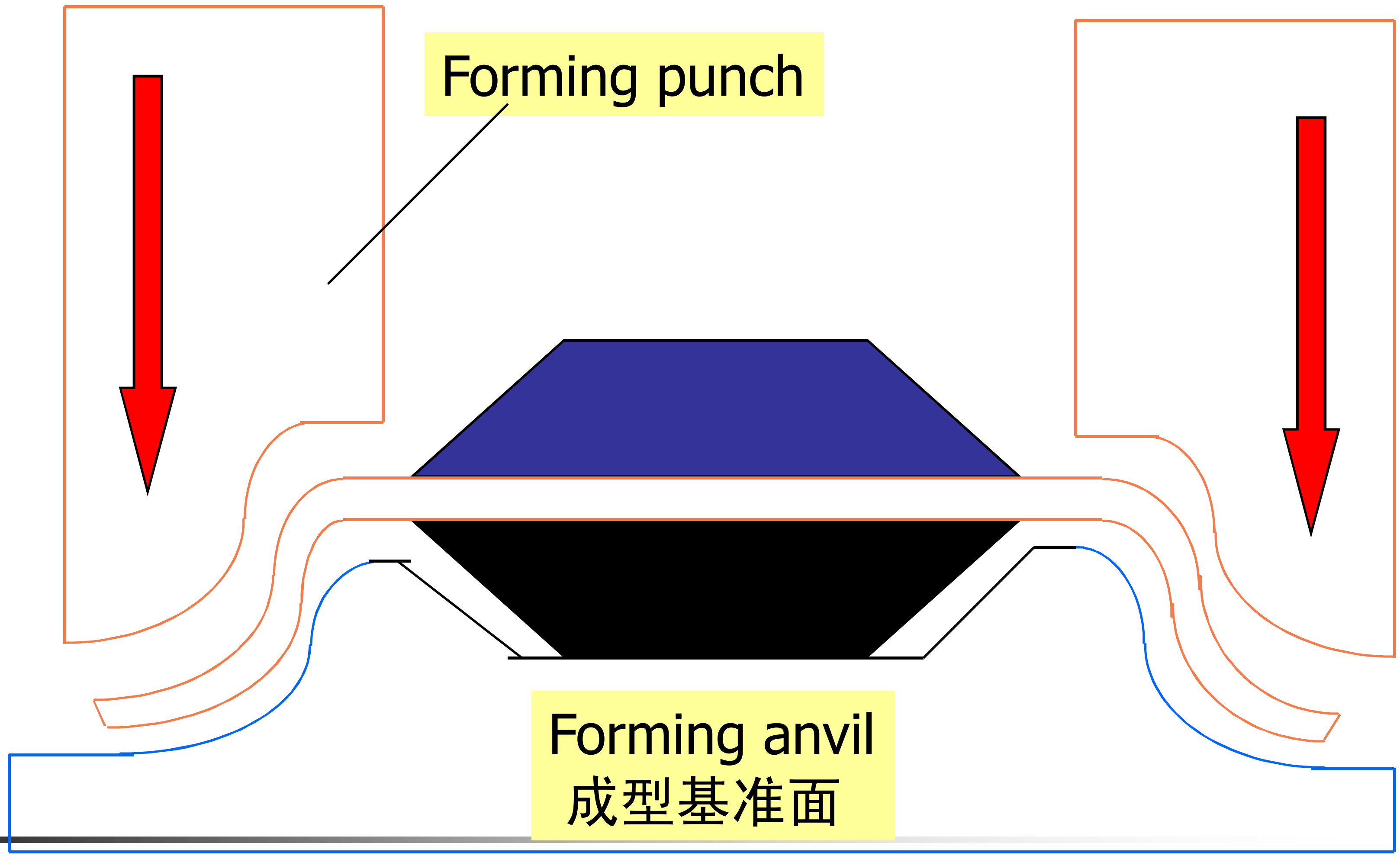
目的：

将已去框（Singulation）Package的Out Lead以连续冲模的方式，将产品引脚弯曲成所要求的形状。





传统IC 成型 Forming



Trim and Form Issues and Challenges

- **Issues**

⇒ **Ease of process**

– **Coplanarity（共面性） control of the finished units.**

– **Solder plating control to avoid excessive thermal shock to the molded body.** 引脚电镀工艺可控

⇒ **Yield**

– **Damaged or missing lead.**

– **Coplanarity reject.**

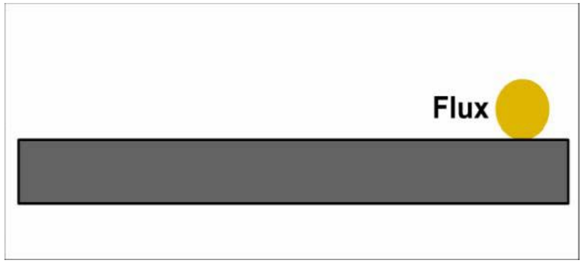
⇒ **Equipment stability and capability**

- **Challenges**

⇒ **Maintaining coplanarity for larger packages.**

2、Ball Attach Process

□ Ball Attach method



（1）真空球粘接方法

- 首先将助焊剂涂覆在操作平台，再用橡胶滚轴将其摊成均匀薄层。
- 接着，将焊球浸入助焊剂，并用真空将焊球吸起并粘接在焊体表面。真空球粘接工具采用网格对准模式将焊球放置在规定位置。
- 特点：助焊剂涂覆焊球表面，使得在回流操作之前焊球可固定在所放置的位置。并且，在回流焊操作时，能防止焊球表面氧化。

2、Ball Attach Process

□ Ball Attach method

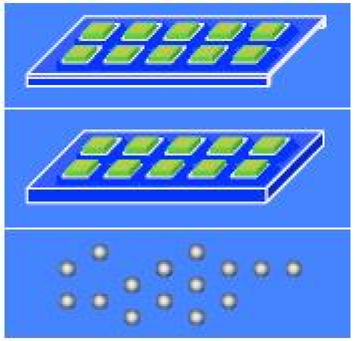
(2)重力球粘接方法

- 首先，助焊剂直接涂覆在盖有焊点掩膜版的封装体表面。
- 接着，焊球被放在一个箱体内，出口具有与焊点掩膜版相同图案的栅孔球罩。
- 助焊剂涂覆以及图形对位完毕后，将箱体倒置，让焊球受重力影响自由下落，直接粘接在封装体对应位置。

2、Ball Attach Process

It is broken down into 8 major steps:

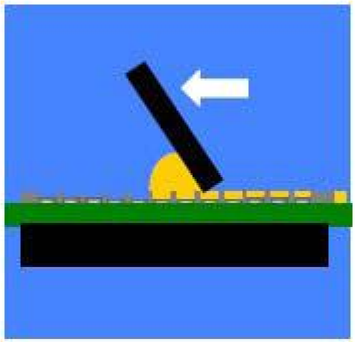
1.Units and Solder Sphere Load (BA)



Objective:

To load the carriers with the units placed on them.To load the ball bin with solder spheres into the machine.

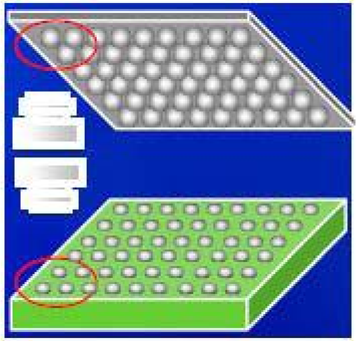
2.Screen Printing



Objective:

To print the flux onto the substrate bump solder.

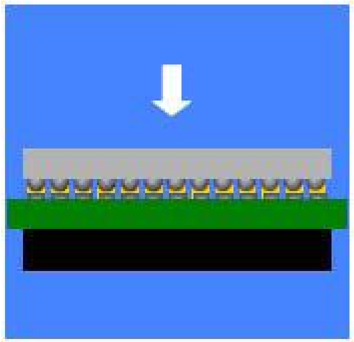
3.Pattern Recognition System (PRS)&Align



Objective:

To align the balls to the theta (rotation) and X-Y position of the units.

4. Ball Placement

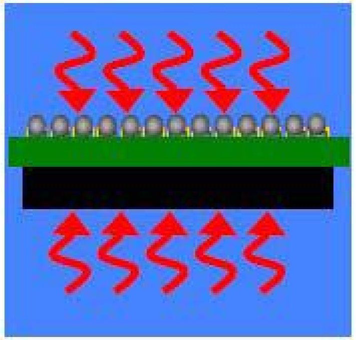


Objective:

To attach the balls onto the substrate.

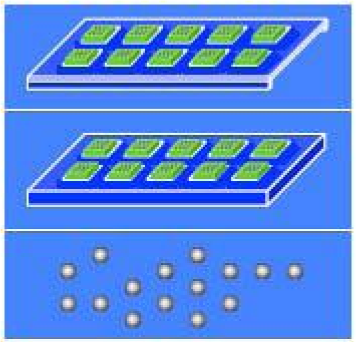
Ball Attach Process (cont.)

5. Reflow



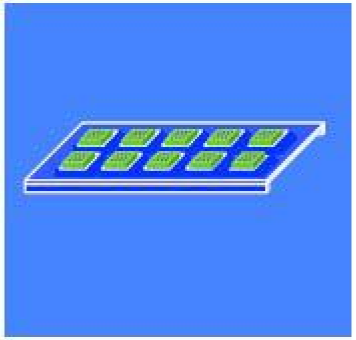
Objective:
To form the interconnection between the die and the substrate.

6.Unload (Die Attach)



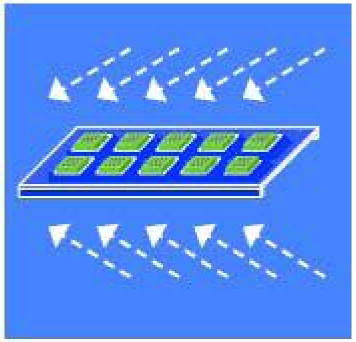
Objective:
To unload the carriers after ball attach. To unload the ball bin when the remaining balls are running out.

7.Load (Deflux)



Objective:
To load the carriers into the deflux machine (if needed).

8. Deflux



Objective:
To remove flux residue in between the balls. Units will be dried after defluxing.

9. Unload (Deflux)

Media and Packing

（传输载体 / 包装）

Media

□ Purpose

MEDIA is used for in-line package assembly processing by carrying usually the singulated parts or components.

□ Media Types:

1)Metal carrier 金属托架

Used to process parts at elevated temperature for a long period of time. Usually used in the front-of-line assembly processes which involves polymer curing at high temperature. Metal carrier is usually more expensive.

Media

2) Plastic tray 塑料托盘

Used to process parts at a lower temperature. Usually used in the end-of-line assembly processes which does not involve processes at high temperature for a long time.

3) Tape and reel 卷带

Used to contain components such as singulated die or capacitors that are assembled onto the substrate.

Packing

- Purpose

PACKING PROCESS is used to protect the finished parts from damage due to moisture, handling and shipment.

- Packing Types:

1) Individual packing (tubes, plastic tray, tape and reel)

--- Units are kept in tubes, plastic tray or tape and reel separating each finished parts from others.

Packing

- Packing Types

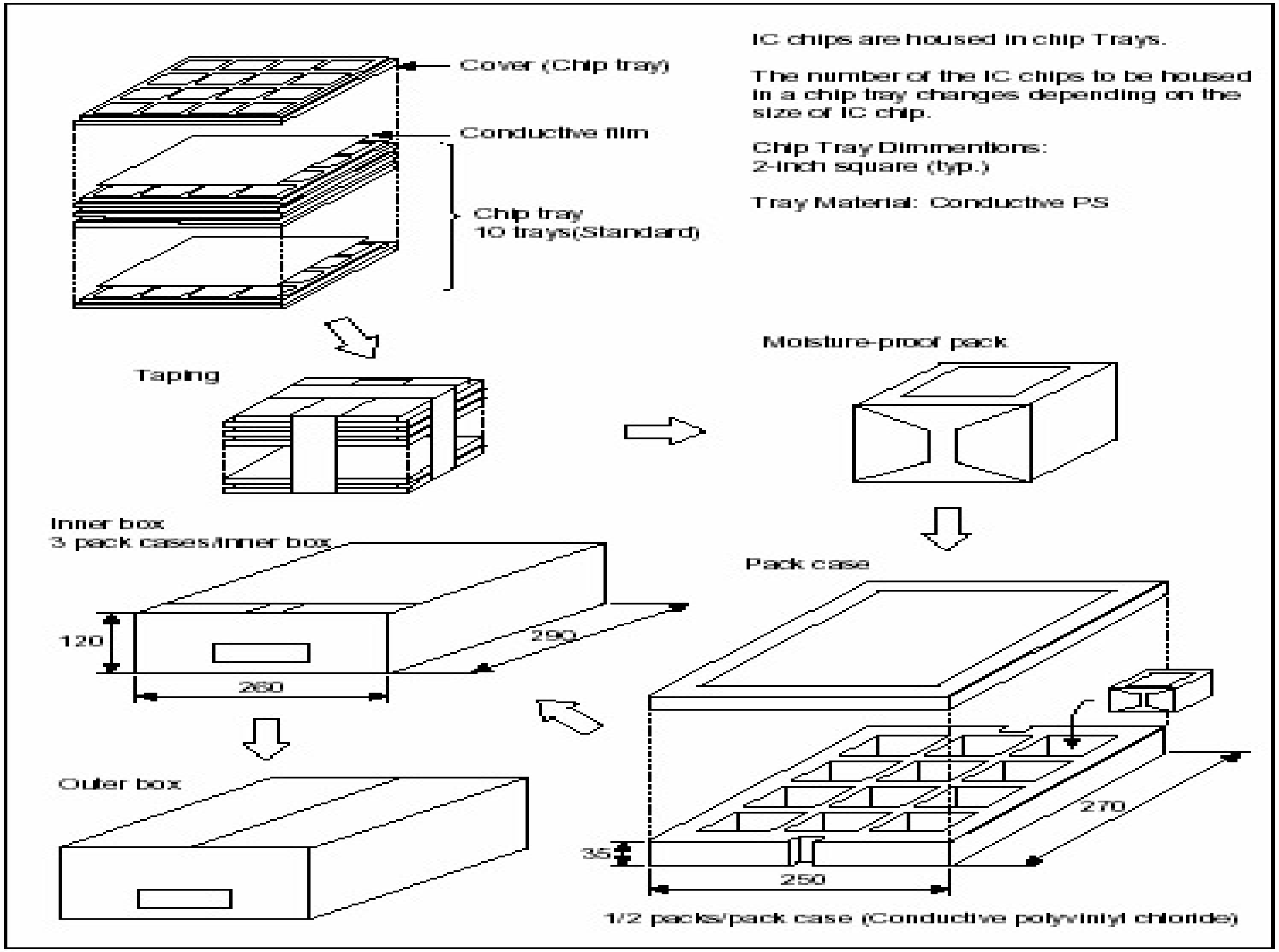
2) Inner box (boxes, empty top and bottom cover trays, cartons, moisture proof bags, with cushioning material)

--- Used to contain the tubes, plastic tray or tape and reel for an easier manual handling.

3) Outer box (bigger boxes with cushioning material)

--- Used to contain the inner boxes for shipment.

Typical packing process



Media and Packing Issues and Challenges

- Issues
  - Ease of process
    - Manual handling
  - Yield
    - Unit drop
  - Equipment stability and capability

- Challenges

  Heavy parts handling.

The End

Q &A